

Chapter 9

Reconfigurable Surfaces

The previous chapters have demonstrated how the signal strength can be increased by equipping the transmitter and receiver with multiple antennas used for precoding and combining. Unfortunately, the MIMO technology can hardly turn a weak channel into a strong one; if the channel gain β is tiny, then $MK\beta$ will remain mediocre. Communication systems that operate under NLOS conditions rely heavily on reflections by various surfaces for the signals to reach the intended receivers. This might lead to multiple propagation paths but often immense signal losses along these paths, particularly in the mmWave and THz bands. Figure 9.1 illustrates such a scenario, where the NLOS receiver can only be reached by beamforming towards a building that reflects the signal. Unfortunately, the building in this example is rotated such that the signal is mainly reflected away from the receiver, following the solid arrow. Can we change the reflection properties somehow so the signal bends around the corner and follows the dashed arrow instead? Yes, this can be achieved using *reconfigurable surfaces*, which is the topic of this chapter.

This chapter will explore how the reflection properties can be dynamically tuned using reconfigurable surfaces to aid the communication between a transmitter and receiver. We begin by explaining how reflections can be interpreted using the beamforming characteristics from previous chapters and how reconfigurable surfaces can control these characteristics. We will then analyze how these surfaces can be configured to maximize the capacity of narrowband and wideband SISO channels and MIMO channels.

9.1 Basic Physics of Reflecting Surfaces

There are two primary categories of reflections: specular and diffuse. These categories are illustrated in Figure 9.2 and represent the extremes in how a plane wave can interact with a reflecting object. In the specular case, the reflected wave remains planar but changes its direction. If angles are measured counterclockwise with 0° being the broadside direction, then a wave with

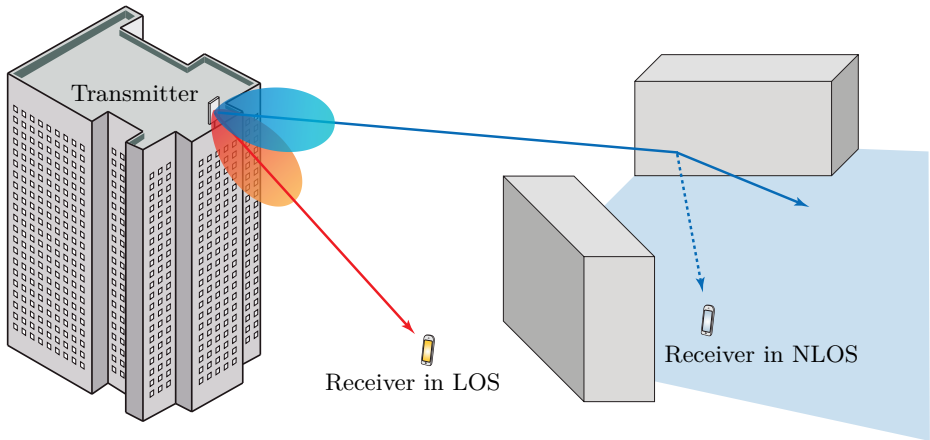


Figure 9.1: A downlink scenario where the transmitter can easily reach the LOS receiver. By contrast, the NLOS receiver has rather weak channel conditions since the wall reflection directs the signal along the solid arrow. If a reconfigurable surface is deployed on that building, the signal can be reflected following the dashed arrow instead.

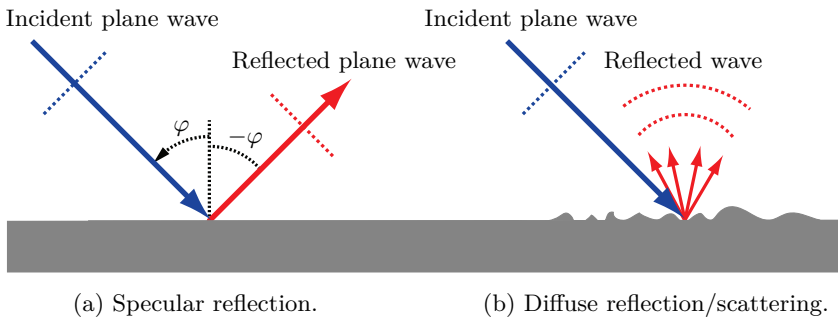


Figure 9.2: A plane wave that reaches an object can be reflected in different ways, with specular reflection and diffuse reflection/scattering being the two extremes.

the incident angle φ is reflecting having the outgoing angle $-\varphi$. This is a consequence of *Snell's law* of refraction, often considered in optics [148]. By contrast, in the diffuse scattering case, the reflected wave has a spherical shape with no particular directivity; thus, the wave's energy is further diffused over the propagation environments.

These categories might seem familiar because we constantly observe how visible light interacts with objects around us to create specular reflections on smooth surfaces (e.g., mirrors that provide an undistorted image) and diffuse reflections on rough surfaces (e.g., white walls that spread the light through the room). The smoothness level is measured compared to the wavelength and size of the object. Firstly, the object must be many wavelengths wide to have the chance to provide (approximately) specular reflection. Secondly, the surface roughness must be small compared to the wavelength. Hence, a large

object that is smooth enough to provide specular reflection for radio waves might be too rough to provide that for visible light. On the other hand, a perfectly smooth but physically small object might be a specular reflector for visible light but be too small to act that way for radio waves since these have a roughly 10^5 times larger wavelength.¹ In fact, the physics that underpins the specular reflection case assumes an infinitely large surface. If a finite-sized mirror approximately provides specular reflection for visible light, it must be 10^5 times larger to give the same approximation accuracy for radio waves. It is not only specular reflection that is an idealization, but ideal diffuse scattering that is uniform in all directions (as in Figure 9.2) is also unlikely to occur in wireless channels; even a rough object has a specific geometry that affects the reflected wave's shape.

The properties of the reflected signal from a finite-sized flat object can be derived using similar methods as in Chapter 4, where we studied antenna arrays. Figure 9.3(a) shows a plane wave impinging on a surface from the angle φ . We will measure the resulting phase-shifts at three points on the surface, which are selected as a ULA with the separation Δ . If we use the left-most point as the phase reference, then the second point observes a phase-shift of $2\pi \frac{\Delta \sin(\varphi)}{\lambda}$ and the third point observes a phase-shift of $2\pi \frac{2\Delta \sin(\varphi)}{\lambda}$. These phases are obtained from the wave needing to travel the additional distances $\Delta \sin(\varphi)$ and $2\Delta \sin(\varphi)$ to reach these points. In general, we obtain the relative phase-shifts at M points in a ULA configuration with separation Δ from the array response vector in (4.19):

$$\mathbf{a}(\varphi) = \begin{bmatrix} 1 \\ e^{-j2\pi \frac{\Delta \sin(\varphi)}{\lambda}} \\ e^{-j2\pi \frac{2\Delta \sin(\varphi)}{\lambda}} \\ \vdots \\ e^{-j2\pi \frac{(M-1)\Delta \sin(\varphi)}{\lambda}} \end{bmatrix} \in \mathbb{C}^M. \quad (9.1)$$

If the same M points retransmit the signal isotropically with the mentioned phase-shifts, we obtain the situation illustrated in Figure 9.3(b). The signal is beamformed using the precoding vector $\mathbf{p} = \mathbf{a}(\varphi)$, which can be expressed as

$$\mathbf{p} = \begin{bmatrix} 1 \\ e^{-j2\pi \frac{\Delta \sin(\varphi)}{\lambda}} \\ e^{-j2\pi \frac{2\Delta \sin(\varphi)}{\lambda}} \\ \vdots \\ e^{-j2\pi \frac{(M-1)\Delta \sin(\varphi)}{\lambda}} \end{bmatrix} = \begin{bmatrix} 1 \\ e^{-j2\pi \frac{\Delta \sin(-\varphi)}{\lambda}} \\ e^{-j2\pi \frac{2\Delta \sin(-\varphi)}{\lambda}} \\ \vdots \\ e^{-j2\pi \frac{(M-1)\Delta \sin(-\varphi)}{\lambda}} \end{bmatrix}^* = \mathbf{a}^*(-\varphi) \quad (9.2)$$

since $\sin(\varphi) = -\sin(-\varphi)$. We recognize this as the MRT vector (without a normalization factor) for transmission in the angular direction $-\varphi$; thus, this

¹This number is obtained by comparing green light having the carrier frequency 600 THz with a wireless communication signal at 6 GHz.

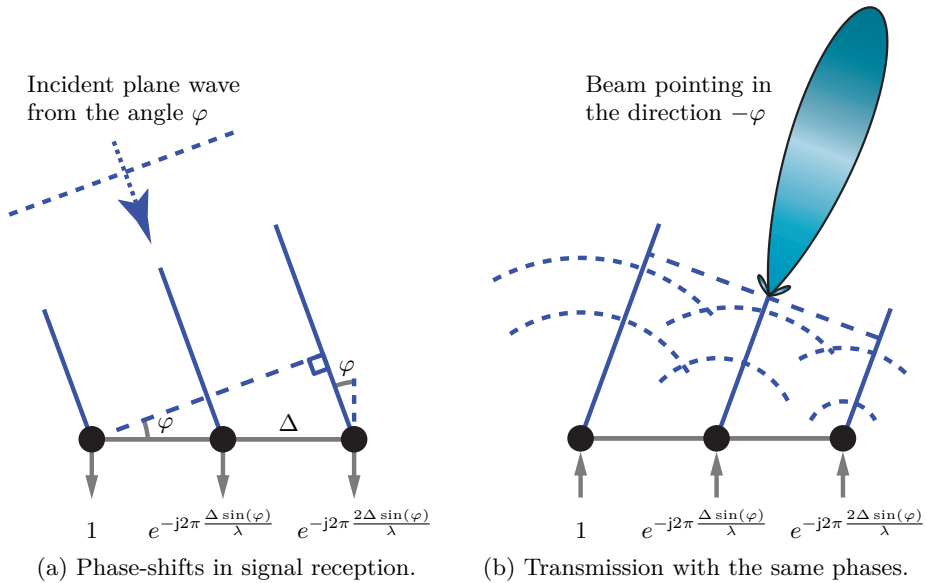


Figure 9.3: When a plane wave impinges on a plane surface from the direction φ , the phase-shifts over the surface can be computed by viewing it as a ULA as in (a). If the same points on the surface retransmit the signal with the phase-shifts obtained from (a), then a beam will be formed in the opposite direction $-\varphi$ as shown in (b).

is where the retransmitted beam is pointing. This observation is essential to determining the shape of the reflected signal from a plane finite-sized surface and is an instance of the *Huygens-Fresnel principle*. This principle says that when a wavefront interacts with an object, every point on that object can be viewed as a new source that emits spherical waves (isotropically). These waves' constructive/destructive combinations determine the new wavefront [149]. The reason that a plane wave that arrives from the angle φ is beamformed with the angle $-\varphi$ is that the distance to a far-away receiver in that direction is identical through all the elements in the ULA shown in Figure 9.3; thus, there will be constructive interference in that direction. The reflected wavefront can be determined using the methodology for computing beam patterns developed in Chapter 4.

9.1.1 Beam Pattern from a Reflecting Surface

We will now compute the angular shape of the reflected signal from the two-dimensional surface illustrated in Figure 9.4, which is deployed in the yz -plane. We denote its horizontal length as L_H and vertical length as L_V to follow the notation from the analysis of UPAs in Section 4.5.3. The considered surface is a homogeneous perfect electric conductor (PEC), but we will first treat it as a UPA by (hypothetically) cutting it into many tiny pieces that each has the

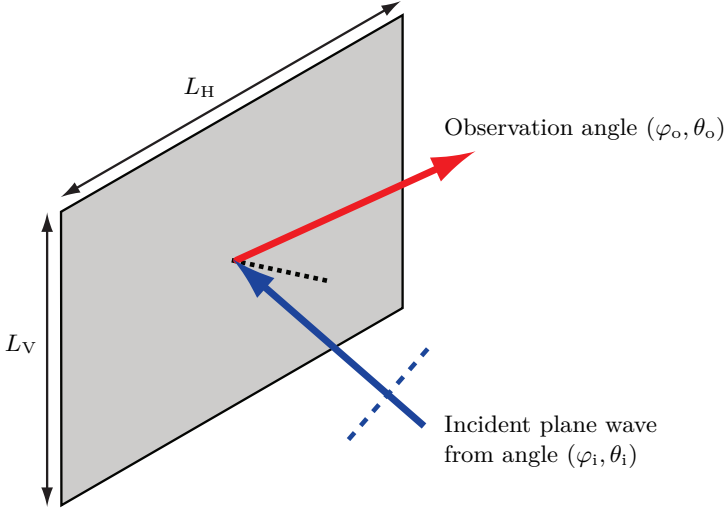


Figure 9.4: A plane wave impinges on a flat homogenous PEC surface in the yz -plane of size $L_H \times L_V$ meters. The channel gain observed in different observation directions (φ_o, θ_o) can be computed similarly to the beam pattern of a UPA, which was characterized in Chapter 4.

physical dimension $\Delta \times \Delta$ for some $\Delta \leq \lambda/4$. Hence, the horizontal/vertical antenna spacing is Δ . Each antenna has an area Δ^2 that is smaller than that of an isotropic antenna, implying that it also has an approximately isotropic radiation pattern. The number of horizontal and vertical antennas can be computed as $N_H = L_H/\Delta$ and $N_V = L_V/\Delta$, respectively. We will use this notation to determine the beam pattern and then let $\Delta \rightarrow 0$ so that the UPA is made of asymptotically many small antennas that we will call *atoms*. The considered setup is shown in Figure 9.4.

If a plane wave impinges on the surface from the azimuth angle $\varphi_i \in [-\pi/2, \pi/2]$ and elevation angle $\theta_i \in [-\pi/2, \pi/2]$, then the relative phase-shifts among the atoms are given by the respective entries of the array response vector $\mathbf{a}_{N_H, N_V}(\varphi_i, \theta_i)$ in (4.128). By following the Huygens-Fresnel principle, we can obtain the reflected signal by considering transmission using the precoding vector

$$\mathbf{p} = \mathbf{a}_{N_H, N_V}(\varphi_i, \theta_i) = \mathbf{a}_{N_H, N_V}^*(-\varphi_i, -\theta_i), \quad (9.3)$$

which corresponds to MRT (without power normalization) in the direction $(-\varphi_i, -\theta_i)$. The beamforming gain that is observed in an arbitrary observation direction, represented by the azimuth angle $\varphi_o \in [-\pi/2, \pi/2]$ and elevation angle $\theta_o \in [-\pi/2, \pi/2]$, is obtained by multiplying with the array response vector $\mathbf{a}_{N_H, N_V}^T(\varphi_o, \theta_o)$ representing the channel in that direction:

$$\begin{aligned} B(\varphi_o, \theta_o) &= |\mathbf{a}_{N_H, N_V}^T(\varphi_o, \theta_o) \mathbf{p}|^2 \\ &= |\mathbf{a}_{N_H, N_V}^T(\varphi_o, \theta_o) \mathbf{a}_{N_H, N_V}^*(-\varphi_i, -\theta_i)|^2. \end{aligned} \quad (9.4)$$

A compact formula for this kind of expression was previously derived in Section 4.5.3, using a slightly different notation. In our case, (4.139) turns into

$$B(\varphi_o, \theta_o) = \frac{\sin^2(\pi L_H \Phi / \lambda)}{\sin^2(\pi \Delta \Phi / \lambda)} \frac{\sin^2(\pi L_V \Omega / \lambda)}{\sin^2(\pi \Delta \Omega / \lambda)}, \quad (9.5)$$

where the impact of the angles is captured by the variables

$$\Phi = \sin(\varphi_o) \cos(\theta_o) + \sin(\varphi_i) \cos(\theta_i), \quad (9.6)$$

$$\Omega = \sin(\theta_o) + \sin(\theta_i). \quad (9.7)$$

Suppose the channel gain from the transmitter to a hypothetical isotropic antenna inside the surface is β_t . Each of the atoms has the (effective) area Δ^2 and will experience the channel gain $\beta_t \Delta^2 / A_{\text{iso}}$ from the transmitter, where $A_{\text{iso}} = \frac{\lambda^2}{4\pi}$ is the area of an isotropic antenna. This is the fraction of the transmitted power that reaches a single atom and will be reflected by it. Similarly, suppose the channel gain from the hypothetical isotropic antenna to a receiver in the observation direction is β_r . Each atom with area Δ^2 will then experience the channel gain $\beta_r \Delta^2 / A_{\text{iso}}$, which is the propagation loss from an atom to the receiver. In conclusion, the end-to-end channel gain from the transmitter to the receiver via the reflecting surface is

$$\begin{aligned} \beta &= \beta_t \frac{\Delta^2}{A_{\text{iso}}} \beta_r \frac{\Delta^2}{A_{\text{iso}}} B(\varphi_o, \theta_o) \\ &= \beta_t \beta_r \frac{\Delta^4}{A_{\text{iso}}^2} \frac{\sin^2(\pi L_H \Phi / \lambda)}{\sin^2(\pi \Delta \Phi / \lambda)} \frac{\sin^2(\pi L_V \Omega / \lambda)}{\sin^2(\pi \Delta \Omega / \lambda)} \\ &\approx \beta_t \beta_r \frac{\Delta^4}{A_{\text{iso}}^2} \frac{\sin^2(\pi L_H \Phi / \lambda)}{(\pi \Delta \Phi / \lambda)^2} \frac{\sin^2(\pi L_V \Omega / \lambda)}{(\pi \Delta \Omega / \lambda)^2} \\ &= \beta_t \beta_r \frac{L_H^2 L_V^2}{A_{\text{iso}}^2} \text{sinc}^2\left(\frac{L_H \Phi}{\lambda}\right) \text{sinc}^2\left(\frac{L_V \Omega}{\lambda}\right), \end{aligned} \quad (9.8)$$

where the approximation utilizes the fact that $\sin(x) \approx x$ when $x \approx 0$ and is tight when $\Delta \rightarrow 0$. The last equality identifies the sinc-function expression. The expression in (9.8) shows how the channel gain depends on the angles and captures all the essential channel properties, except for polarization. The potential polarization mismatch between the transmitter and receiver can be included in β_t and β_r but are also angle-dependent [150].

The largest channel gain is obtained in (9.8) when $\Phi = \Omega = 0$. By inspecting (9.6) and (9.7), we can conclude that the maximum is obtained for the observation angles $\varphi_o = -\varphi_i$ and $\theta_o = -\theta_i$, which is expected from Snell's law and the previous discussion. If the reflected signal were a plane wave, the channel gain would be zero in all other directions, which is not the case. Instead, the angular gain variations are the same as for a UPA with the same physical size and can be analyzed as in Section 4.5.3.

Example 9.1. What are the reflected signal's first-null horizontal and vertical beamwidths when a plane wave impinges from $\varphi_i = \theta_i = 0$?

The first nulls appear when the argument of one of the sinc-functions in (9.8) is ± 1 . In the horizontal plane (i.e., $\theta_o = 0$), this happens for $\varphi_o = \pm \arcsin(\lambda/L_H) \approx \lambda/L_H$. Hence, the first-null horizontal beamwidth is approximately $2\lambda/L_H$. It follows from the same computation that the first-null vertical beamwidth is approximately $2\lambda/L_V$. The beamwidths are proportional to the wavelength, which demonstrates how a PEC surface of a given physical size can give an extremely narrow beamwidth for visible light but a relatively wide beamwidth for radio spectrum.

As the surface's lengths L_H, L_V grow large, for a given wavelength, the beamwidths approach zero. This implies the reflected signal will be a plane wave with zero beamwidth in the asymptotic limit. This corresponds to ideal specular reflection where the incident plane wave only changes direction.

The maximum gain value in (9.8) can be factorized as

$$\beta = \beta_t \cdot \underbrace{\frac{L_H L_V}{A_{\text{iso}}}}_{=\text{Aperture gain for reception}} \cdot \beta_r \cdot \underbrace{\frac{L_H L_V}{A_{\text{iso}}}}_{=\text{Beamforming gain for retransmission}}. \quad (9.9)$$

Recall that $L_H L_V$ is the total area of the surface. The first term in (9.9) is the channel gain from the transmitter to an isotropic-antenna-sized receiver surface, while the second term $L_H L_V / A_{\text{iso}}$ determines how much larger aperture the surface has. Hence, we will call the second term the *aperture gain*, but it could also be interpreted as a receive beamforming gain. This part of the expression models the fact that a surface collects an amount of power from the impinging plane wave that is proportional to its area. The third term in (9.9) is the channel gain from an isotropic-antenna-sized transmitter surface to the receiver, while the fourth term $L_H L_V / A_{\text{iso}}$ determines the transmit *beamforming gain* delivered by the surface. This part of the expression highlights how a big surface can beamform/reflect the signal with a narrower beamwidth and a power concentration in the main beam proportional to its area.

Figure 9.5 shows the end-to-end channel gain in (9.8) observed in different angle directions φ_o in the azimuth plane (where $\theta_o = 0$) when a plane wave impinges from the direction $\varphi_i = \pi/6$, $\theta_i = 0$. We consider a square surface with the side lengths $L = L_H = L_V \in \{4\lambda, 16\lambda\}$ and a propagation scenario with $\beta_t \beta_r = 10^{-8}$. The figure shows how the reflected signals are beams pointing in the direction $\varphi_o = -\pi/6 = -\varphi_i$, as expected. The horizontal beamwidth shrinks slightly when the surface increases in size, but the more dominant effect is the increased channel gain, which grows quadratically with the surface area. Hence, when each side increases by a factor of 4, the channel gain grows by $4^4 = 24$ dB. This is the combination of the aperture gain and the transmit beamforming gain.

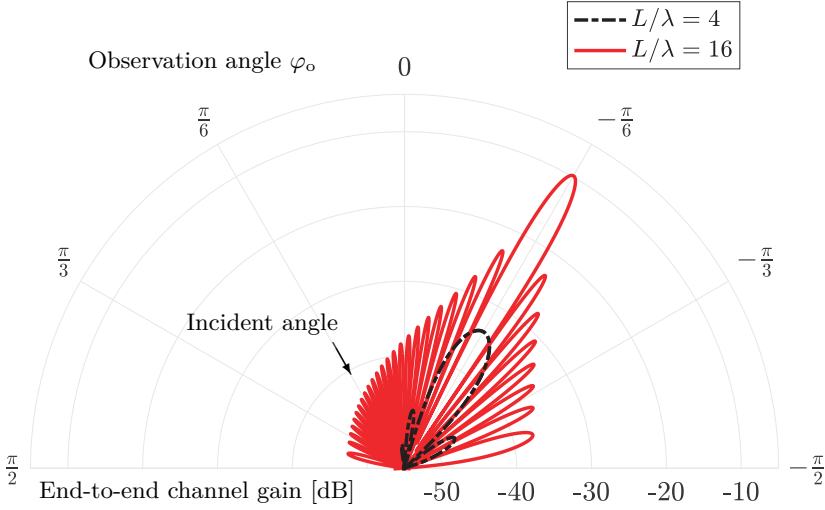


Figure 9.5: The end-to-end channel gain in (9.8) from a transmitter in the direction $\varphi_i = \pi/6$ via a reflecting PEC surface to a receiver in a varying observation angle direction φ_o . The channel gain depends on the surface's size $L \times L$ and the observation angle.

Example 9.2. How does the end-to-end channel gain in (9.8) relate to the radar range equation in (8.71)?

The connection between these expressions can be identified by using the notation $\beta_t = G_t(\varphi_t, \theta_t) \frac{\lambda^2}{(4\pi d_t)^2}$ and $\beta_r = G_r(\varphi_r, \theta_r) \frac{\lambda^2}{(4\pi d_r)^2}$ for the channel gains of the LOS paths to and from the radar target, respectively. These channel gains depend on the propagation distances and antenna gains at the transmitter and receiver. The received power in (8.71) can then be expressed as

$$P_r = P_t \beta_t \beta_r \frac{\sigma_{\text{RCS}}}{A_{\text{iso}}}. \quad (9.10)$$

The corresponding received power when using the reflecting surface is $P_t \beta$ so by comparing (9.10) with (9.8), we can identify the RCS of the surface as

$$\sigma_{\text{RCS}} = \frac{L_H^2 L_V^2}{A_{\text{iso}}} \text{sinc}^2 \left(\frac{L_H \Phi}{\lambda} \right) \text{sinc}^2 \left(\frac{L_V \Omega}{\lambda} \right). \quad (9.11)$$

This expression characterizes how the RCS depends on the incident and observation angles through Φ and Ω . The largest value is achieved when $\Phi = \Omega = 0$, but we can also achieve zero RCS if $\Phi = \pm \lambda/L_H$ or $\Omega = \pm \lambda/L_V$.

If the surface would be rotated randomly with respect to the transmitter and receiver, then Φ and Ω are random, and so is the angle-dependent RCS σ_{RCS} . This is the core principle that leads to randomness in radar sensing.

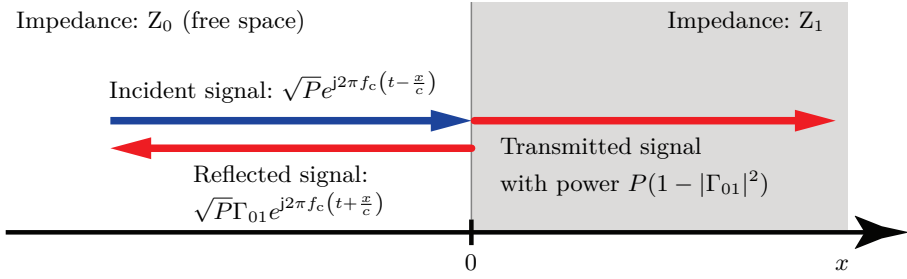


Figure 9.6: When the plane wave $\sqrt{P}e^{j2\pi f_c(t - \frac{x}{c})}$ reaches the boundary between two mediums, a fraction of it is reflected backward and the remaining is transmitted into the new medium. The reflection coefficient Γ_{01} determines these behaviors.

9.1.2 Reconfigurable Reflection from Heterogeneous Surfaces

The previous section analyzed the reflection from a homogeneous surface, which reflects all the power of the incident wave as a beam pointing in the specular reflection direction. The situation is different when the surface is heterogeneous. We need the *reflection coefficient* to study that scenario.

When a sinusoidal wave reaches the boundary between two mediums, their respective characteristic impedances determine what fraction of the signal is reflected back versus transmitted into the new medium. We let Z_0 denote the impedance of the first medium (e.g., free space) and Z_1 denote the impedance of the second medium (e.g., the surface). The reflection coefficient can then be computed as [151, Sec. 1.7]

$$\Gamma_{01} = \frac{Z_1 - Z_0}{Z_1 + Z_0} \quad (9.12)$$

and is the reflected signal divided by the incident signal at the boundary between the mediums. The reflection coefficient can be complex, in which case $\arg(\Gamma_{01})$ represents the phase-shift incurred to the signal before it is reflected. This scenario is illustrated in Figure 9.6, where the first medium is free space (vacuum) for which the speed of light has been denoted c earlier in this book. A fraction $|\Gamma_{01}|^2$ of the power is reflected, while the remaining fraction $1 - |\Gamma_{01}|^2$ is transmitted into the new medium and might be absorbed by it. We will focus on the reflected signal.

A homogenous surface has a constant impedance Z_1 , which results in a reduced power by a factor of $|\Gamma_{01}|^2 \in [0, 1]$, but otherwise, the same reflection behavior as in the previous section. However, suppose the surface is divided into N small units that are structurally similar but have heterogeneous electrical properties. We call these *metaatoms* and each has a specific impedance Z_n for $n = 1, \dots, N$. The corresponding reflection coefficients then become

$$\Gamma_{0n} = \frac{Z_n - Z_0}{Z_n + Z_0}. \quad (9.13)$$

The characteristic impedance of free space is $Z_0 \approx 376.7 \approx 120\pi$ ohm, so it is real-valued. Suppose we design the metaatom using a reactance element with a purely imaginary impedance $Z_n = jX_n$ for some $X_n \in \mathbb{R}$. It then follows that

$$|\Gamma_{0n}| = \left| \frac{jX_n - Z_0}{jX_n + Z_0} \right| = \frac{\sqrt{X_n^2 + Z_0^2}}{\sqrt{X_n^2 + Z_0^2}} = 1, \quad (9.14)$$

$$\arg(\Gamma_{0n}) = \arg\left(\frac{jX_n - Z_0}{jX_n + Z_0}\right) = \begin{cases} \pi - 2 \arctan\left(\frac{X_n}{Z_0}\right), & \text{if } X_n \geq 0, \\ -\pi - 2 \arctan\left(\frac{X_n}{Z_0}\right), & \text{if } X_n < 0. \end{cases} \quad (9.15)$$

Such a metaatom will reflect all the incident power since $|\Gamma_{0n}| = 1$ and it causes a phase-shift $\psi_n = \arg(\Gamma_{0n})$ that can be continuously tuned between $-\pi$ and π by varying X_n . Such tuning can be achieved by configuring a capacitor that determines the capacitive part of the reactance, which can be implemented using a varactor diode.² This feature is the key to designing reconfigurable surfaces that can shape the reflected signals.

We will now return to the reflection example in Figure 9.4 that analyzed a homogeneous PEC surface. Suppose that surface is replaced by the one shown in Figure 9.7, which consists of $N = N_H N_V$ metaatoms that have varying impedance values corresponding to phase-shifts between $-\pi$ and π . Each color represents a specific phase value, and we let $\psi_n \in [-\pi, \pi)$ denote the phase-shift incurred by the n th metaatom. If the incident plane wave arrives from the angular direction (φ_i, θ_i) , then the incident phase-shifts over the surface are given by the array response vector $\mathbf{a}_{N_H, N_V}(\varphi_i, \theta_i)$. Each metaatom then adjusts its local incident phase value by ψ_n ; therefore, the phase profile of the retransmitted/reflected signals is given by the precoding vector

$$\mathbf{p} = \mathbf{D}_\psi \mathbf{a}_{N_H, N_V}(\varphi_i, \theta_i) = \mathbf{D}_\psi \mathbf{a}_{N_H, N_V}^*(-\varphi_i, -\theta_i), \quad (9.16)$$

where the surface's phase adjustments are applied using the diagonal *reflection matrix*

$$\mathbf{D}_\psi = \text{diag}\left(e^{j\psi_1}, \dots, e^{j\psi_N}\right). \quad (9.17)$$

The beamforming gain expression in (9.4) can then be updated as

$$\begin{aligned} B(\varphi_o, \theta_o) &= |\mathbf{a}_{N_H, N_V}^T(\varphi_o, \theta_o) \mathbf{p}|^2 \\ &= |\mathbf{a}_{N_H, N_V}^T(\varphi_o, \theta_o) \mathbf{D}_\psi \mathbf{a}_{N_H, N_V}^*(-\varphi_i, -\theta_i)|^2. \end{aligned} \quad (9.18)$$

Since each entry of an array response vector is a complex exponential entirely determined by a phase value, we can turn $\mathbf{p}$ into any array response vector of our choice by selecting $\mathbf{D}_\psi$ accordingly. We can thereby control the main direction of the reflected beam. We can also generate precoding vectors that are not array response vectors if we happen to prefer that.

²A capacitor adds a positive value to X_n and an inductor adds a negative value. If the metaatom is a circuit consisting of both fixed inductive elements and variable capacitive elements, then we can control X_n over a range of both positive and negative values, resulting in the range of positive and negative phase-shifts shown in Figure 9.17.

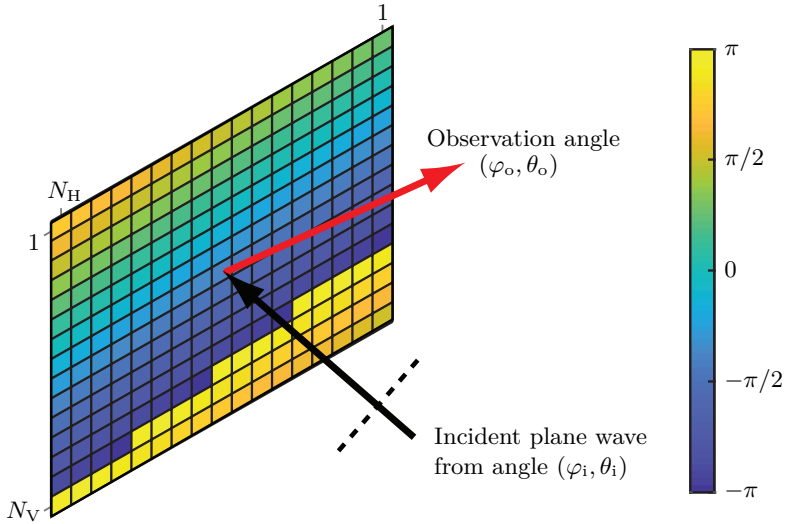


Figure 9.7: A plane wave impinges on a flat surface in the yz -plane that consists of $N_H \times N_V$ reflecting metaatoms with heterogeneous properties. The metaatoms' varying impedances cause different phase-shifts between $-\pi$ and π , as exemplified using colors. This enables the surface to control the channel gain in the observation directions (φ_o, θ_o) .

Example 9.3. How should the surface's phase-shifts be selected to point the reflected beam in a specific desired direction (φ_d, θ_d) ?

The reflected beam will point in that direction if $\mathbf{p} = \mathbf{a}_{N_H, N_V}^*(\varphi_d, \theta_d)$. By equating (9.16) to this value, we obtain the relation

$$\mathbf{D}_\psi \underbrace{\mathbf{a}_{N_H, N_V}^*(-\varphi_i, -\theta_i)}_{=[a_{i,1}, \dots, a_{i,N}]^T} = \underbrace{\mathbf{a}_{N_H, N_V}^*(\varphi_d, \theta_d)}_{=[a_{d,1}, \dots, a_{d,N}]^T}. \quad (9.19)$$

The n th entry can be expressed as $e^{j\psi_n} a_{i,n} = a_{d,n}$, which holds if $\psi_n = \arg(a_{d,n}/a_{i,n})$. Hence, each metaatom compensates for the phase difference between the desired array response and the actual array response of the incident wave. Since the phase varies gradually in both vectors, the phase profile of the surface will also vary gradually, as exemplified in Figure 9.7.

Another way to control the direction of the reflected beam would be to rotate the surface mechanically, but greater flexibility is achieved by the electrical implementation described above. A similar discussion was made in relation to Example 4.22, which compared the mechanical and electrical downtilt of an antenna array. When considering reflecting surfaces, we seek a way to deploy them on building facades, as illustrated in Figure 9.1, to point the reflection angle toward the receiving user without mechanical rotations. Using the radar sensing terminology from Section 8.3, we want to configure the electrical properties of the surface to achieve the largest possible RCS in the direction leading to the receiver.

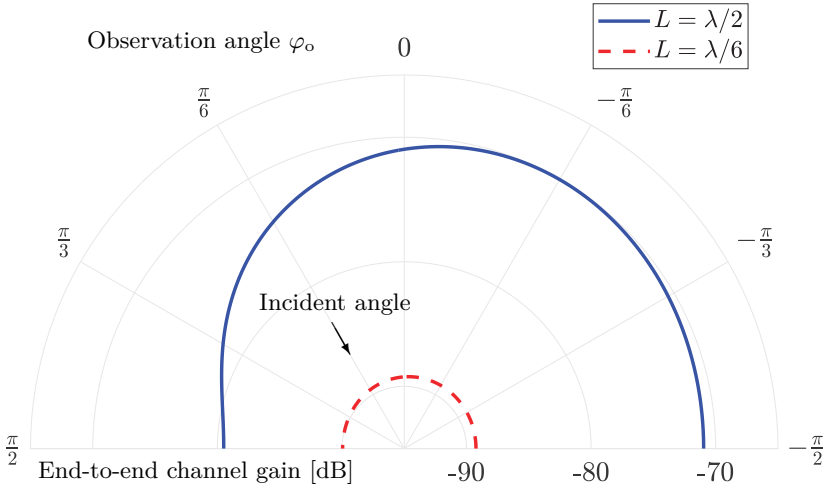


Figure 9.8: The end-to-end channel gain in (9.8) from a transmitter in the direction $\varphi_i = \pi/6$ via a reflecting surface of size $L \times L$ to a receiver in a varying observation angle direction φ_o . The smaller the reflector is, the more circular/isotropic its radiation pattern is.

We use the term “metaatom” when referring to each controllable piece of the surface to signify that they are tiny compared to the wavelength. This is because a small object provides approximately diffuse reflection with no preferred directivity, even if it is flat. Figure 9.8 shows the end-to-end channel gain in (9.8) observed in different azimuth angle directions φ_o when a plane wave impinges from the direction $\varphi_i = \pi/6$, $\theta_i = 0$. The setup is the same as in Figure 9.5, but now the surface dimensions are $L \times L$ with $L \in \{\lambda/6, \lambda/2\}$. Both sizes result in a reflected signal that is spread over all angles, even back toward the transmitter, but the radiation pattern becomes more circular (i.e., closer to isotropic) as the size shrinks. For the reconfigurable surface to fully steer the direction of the reflected signal, it should be made of many tiny controllable pieces that each lack a preferable directivity but can be used to jointly beamform the reflected signal where we want it to go.

9.1.3 Terminology and Implementation Aspects

Reconfigurable surfaces are often associated with metamaterials, which are engineered materials typically containing sub-wavelength-sized structures that create a heterogeneous impedance profile over the surface. These structures are typically referred to as metaatoms, which is why we have already adopted that terminology in this chapter. The engineered material concept was first utilized in communications to design static *reflectarrays* with a fixed reflection matrix $\mathbf{D}_\psi$ that was not a scaled identity matrix. This results in an anomalous reflection angle that differs from Snell’s law [152]. This was followed by

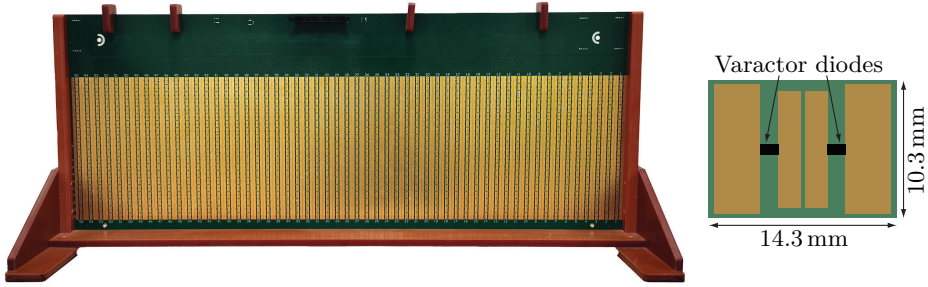


Figure 9.9: The photo shows the reconfigurable surface prototype from [159], which is designed for the 5.8 GHz band. It consists of $N = 1100$ metaatoms, arranged as a UPA with $N_V = 20$ rows and $N_H = 55$ metaatoms per row. The impedance of each metaatom is controlled using two varactor diodes, which enable phase control of the reflected signals over a range of 240° .

reconfigurable reflectarrays that can electrically tune the reflection matrix [153]. The purpose was to build transmitters/receivers consisting of a single antenna pointing towards the reflectarray that controls the beam direction. This is an alternative implementation of the analog beamforming architecture, discussed in Section 7.3.1, that is particularly used for satellite communications and radars but has also been commercialized for mmWave transceivers.

The alternative concept of deploying reconfigurable surfaces in the propagation environment to relay signals between a transmitter and receiver gained traction in the late 2010s. This concept has been called *software-controlled metasurfaces* [154], *reconfigurable intelligent surfaces (RIS)* [155], *intelligent reflecting surfaces (IRS)* [156], and *reconfigurable intelligent metasurfaces* [157]. In this book, we will call them reconfigurable surfaces. There is a wealth of implementation challenges and details that go beyond the purpose of this book. We refer to [158] for a review of software-controlled metasurfaces designed for everything from the low-band to the infrared frequency range. While it is possible to build reconfigurable surfaces that can vary the phase-shifts continuously using varactor diodes, many designs use PIN diodes that can be switched on and off to shift between a discrete set of phase values.

A reconfigurable surface prototype from [159] with $N = 1100$ metaatoms is shown in Figure 9.9. Each metaatom contains two metallic patches connected to varactor diodes, which are controlled by an external bias voltage to tune the impedance. This enables the prototype to select phases $\psi_n \in [-120^\circ, 120^\circ]$ and the corresponding power loss varies slightly with the phase but satisfies $1 - |\Gamma_{0n}|^2 < -3\text{ dB}$. The indoor and outdoor measurements presented in [159] verify that the prototype can change the angle of the reflected beam as described above. In conclusion, reconfigurable surfaces can be implemented, and the remainder of this chapter will analyze how they can aid communication and radar systems.

9.2 Narrowband Communication using Reconfigurable Surfaces

We will now analyze how a reconfigurable surface can be tuned to aid a point-to-point communication system. We begin by considering a narrowband SISO channel between a single-antenna transmitter and a single-antenna receiver. The received signal was stated in (2.144) as

$$y = h \cdot x + n, \quad (9.20)$$

where h is the channel coefficient, $x \sim \mathcal{N}_{\mathbb{C}}(0, q)$ is the capacity-achieving transmit signal, and $n \sim \mathcal{N}_{\mathbb{C}}(0, N_0)$ is independent noise. According to Corollary 2.1, the capacity of such a channel can be expressed as

$$C = \log_2 \left(1 + \frac{q|h|^2}{N_0} \right) \quad \text{bit/symbol.} \quad (9.21)$$

When a reconfigurable surface is deployed in the propagation environment, it affects how the channel coefficient h is modeled. Suppose the surface consists of N metaatoms that reflect all the incident power with the controllable phase-shifts ψ_n for $n = 1, \dots, N$. This setup is illustrated in Figure 9.10. In general, the end-to-end channel can be modeled as

$$h = h_s + \sum_{n=1}^N h_{r,n} e^{j\psi_n} h_{t,n}, \quad (9.22)$$

where the *static channel* $h_s \in \mathbb{C}$ includes all propagation paths unaffected by the surface. The propagation path via metaatom n is described by the channel coefficient $h_{t,n} \in \mathbb{C}$ from the transmitter to the metaatom, the phase-shift $e^{j\psi_n}$ incurred by the metaatom, and the channel coefficient $h_{r,n} \in \mathbb{C}$ from the metaatom to the receiver. These three coefficients are multiplied together following Section 9.1.1. Since waves that travel through different paths are superimposed at the receive antenna, the channel coefficients are added up as in (9.22). We can express (9.22) in the matrix/vector form

$$h = h_s + \mathbf{h}_r^T \mathbf{D}_\psi \mathbf{h}_t \quad (9.23)$$

by introducing the notation

$$\mathbf{h}_t = \begin{bmatrix} h_{t,1} \\ \vdots \\ h_{t,N} \end{bmatrix}, \quad \mathbf{h}_r = \begin{bmatrix} h_{r,1} \\ \vdots \\ h_{r,N} \end{bmatrix}, \quad (9.24)$$

and recalling the reflection matrix notation $\mathbf{D}_\psi = \text{diag}(e^{j\psi_1}, \dots, e^{j\psi_N})$ from (9.17). Using this notation, the capacity in (9.21) for a given value of $\mathbf{D}_\psi$ becomes

$$\log_2 \left(1 + \frac{q|h_s + \mathbf{h}_r^T \mathbf{D}_\psi \mathbf{h}_t|^2}{N_0} \right). \quad (9.25)$$

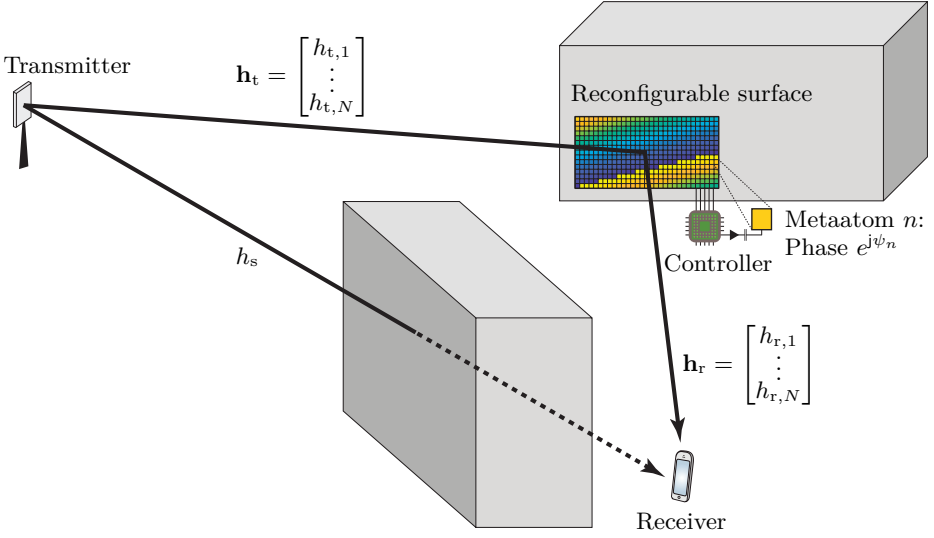


Figure 9.10: An example of a narrowband SISO channel aided by a reconfigurable surface with N metaatoms. The SIMO channel from the single-antenna transmitter to the surface is denoted by $\mathbf{h}_t$ and the MISO channel from the surface to the single-antenna receiver is denoted by $\mathbf{h}_r$. Each metaatom incurs a tunable phase-shift $e^{j\psi_n}$ and the static channel that does not involve the surface is denoted as h_s . The end-to-end channel coefficient becomes $h = h_s + \mathbf{h}_r^T \mathbf{D}_\psi \mathbf{h}_t$, where $\mathbf{D}_\psi = \text{diag}(e^{j\psi_1}, \dots, e^{j\psi_N})$ is the reflection matrix.

We can aid the system by identifying the matrix that maximizes this capacity expression. In particular, we want to maximize

$$\begin{aligned}
 |h_s + \mathbf{h}_r^T \mathbf{D}_\psi \mathbf{h}_t|^2 &= \left\| \begin{bmatrix} \sqrt{|h_s|} \\ \sqrt{|h_{r,1} h_{t,1}|} \\ \vdots \\ \sqrt{|h_{r,N} h_{t,N}|} \end{bmatrix} \right\|^H \left\| \begin{bmatrix} \sqrt{|h_s|} e^{j \arg(h_s)} \\ \sqrt{|h_{r,1} h_{t,1}|} e^{j(\arg(h_{r,1} h_{t,1}) + \psi_1)} \\ \vdots \\ \sqrt{|h_{r,N} h_{t,N}|} e^{j(\arg(h_{r,N} h_{t,N}) + \psi_N)} \end{bmatrix} \right\|^2 \\
 &\leq \left\| \begin{bmatrix} \sqrt{|h_s|} \\ \sqrt{|h_{r,1} h_{t,1}|} \\ \vdots \\ \sqrt{|h_{r,N} h_{t,N}|} \end{bmatrix} \right\|^2 \left\| \begin{bmatrix} \sqrt{|h_s|} e^{j \arg(h_s)} \\ \sqrt{|h_{r,1} h_{t,1}|} e^{j(\arg(h_{r,1} h_{t,1}) + \psi_1)} \\ \vdots \\ \sqrt{|h_{r,N} h_{t,N}|} e^{j(\arg(h_{r,N} h_{t,N}) + \psi_N)} \end{bmatrix} \right\|^2 \\
 &= \left(|h_s| + \sum_{n=1}^N |h_{r,n} h_{t,n}| \right)^2 \quad (9.26)
 \end{aligned}$$

where the second row follows from the Cauchy-Schwartz inequality in (2.18). The upper bound in that inequality is achieved if and only if the two vectors are equal, except for a scaling factor. The first entries of the two vectors differ by the phase-shift $e^{j \arg(h_s)}$, which is determined by the static channel

coefficient and cannot be changed. The other entries differ by a phase-shift that the surface can control; thus, we can attain the upper bound by matching their phases to the first entry. In particular, the entry corresponding to metaatom n must be configured such that

$$e^{j\arg(h_{r,n}h_{t,n})}e^{j\psi_n} = e^{j\arg(h_s)} \Rightarrow \psi_n = \arg(h_s) - \arg(h_{r,n}h_{t,n}) + 2\pi k_n \quad (9.27)$$

for the integer k_n that gives that $\psi_n \in [-\pi, \pi)$. This solution is also obtained as $\psi_n = \arg(h_s/(h_{r,n}h_{t,n}))$ if $h_s \neq 0$ and $h_{r,n}h_{t,n} \neq 0$. We notice that the capacity-maximizing configuration removes the phase-shift $\arg(h_{r,n}h_{t,n})$ created by the channels to and from the metaatom and replaces it with the phase-shift $\arg(h_s)$ of the static channel. In this way, the signals that the N metaatoms reflect reach the receiver with a phase that matches the signal that propagates through the static channel. We have proved the following result.

Corollary 9.1. Consider a discrete memoryless SISO channel aided by a reconfigurable surface, for which the channel coefficient is

$$h = h_s + \sum_{n=1}^N h_{r,n}e^{j\psi_n}h_{t,n}. \quad (9.28)$$

The channel capacity is maximized by configuring the surface as $\psi_n = \arg(h_s) - \arg(h_{r,n}h_{t,n}) + 2\pi k_n$, where k_n is the integer that gives $\psi_n \in [-\pi, \pi)$, for $n = 1, \dots, N$. This results in the capacity

$$C = \log_2 \left(1 + \frac{q \left(|h_s| + \sum_{n=1}^N |h_{r,n}h_{t,n}| \right)^2}{N_0} \right) \quad \text{bit/symbol.} \quad (9.29)$$

The maximum end-to-end channel gain is the squared sum of the amplitudes $|h_s|$ and $|h_{r,n}h_{t,n}|$ for $n = 1, \dots, N$. If we define the *effective channel vector*

$$\check{\mathbf{h}} = \begin{bmatrix} h_s \\ h_{r,1}h_{t,1} \\ \vdots \\ h_{r,N}h_{t,N} \end{bmatrix}, \quad (9.30)$$

we can alternatively express the end-to-end channel gain using the 1-norm³ as

$$\left(|h_s| + \sum_{n=1}^N |h_{r,n}h_{t,n}| \right)^2 = \|\check{\mathbf{h}}\|_1^2. \quad (9.31)$$

³The 1-norm is defined for an arbitrary vector $\mathbf{x} \in \mathbb{C}^M$ as $\|\mathbf{x}\|_1 = \sum_{m=1}^M |x_m|$. It is also known as the Manhattan norm since it adds up the distances in the M dimensions as if one has to travel along straight perpendicular streets on a map.

Example 9.4. Suppose the N reflected paths have the same propagation losses: $|h_{t,n}|^2 = \beta_t$ and $|h_{r,n}|^2 = \beta_r$ for $n = 1, \dots, N$. How does the end-to-end channel gain behave when the channel gain $|h_s|^2 = \beta_s$ of the static path is either relatively weak or strong?

The contribution from the metaatoms to (9.29) is $\sum_{n=1}^N |h_{r,n} h_{t,n}| = N\sqrt{\beta_r \beta_t}$ under these assumptions. Hence, the end-to-end channel gain becomes

$$\left(\sqrt{\beta_s} + N\sqrt{\beta_r \beta_t}\right)^2 \approx \begin{cases} N^2 \beta_r \beta_t, & \text{if } \beta_s \text{ is small,} \\ \beta_s, & \text{if } \beta_s \text{ is large,} \end{cases} \quad (9.32)$$

where “small” means that $\beta_s \ll N^2 \beta_r \beta_t$ and “large” means that $\beta_s \gg N^2 \beta_r \beta_t$. In the former case, when the vast majority of the received power comes from the surface, the end-to-end channel gain is proportional to $N^2 \beta_r \beta_t$. This term grows quadratically with the number of metaatoms, thanks to an aperture gain of N and a transmit beamforming gain of N . When the static path is relatively strong (i.e., $\beta_s \gg N^2 \beta_r \beta_t$), the reconfigurable surface barely affects the end-to-end channel gain, which is approximately equal to β_s .

The conclusion is that physically large reconfigurable surfaces are much more effective than small surfaces, thanks to the quadratic scaling law. This is important because the N^2 factor is multiplied by $\beta_r \beta_t$, which is the product of two channel gains that both can be very small numbers.

We will now use the exact expression on the left-hand side of (9.32) to demonstrate how the number of metaatoms affects communication performance. Figure 9.11 shows the capacity as a function of the number of metaatoms when $\beta_t = -80$ dB, $\beta_r = -60$ dB, and $q/N_0 = 100$ dB. We compare two types of static paths: $\beta_s = -80$ dB (strong) and $\beta_s = -110$ dB (weak). When the static path is weak, the capacity is nearly zero for $N = 0$ but grows rapidly as metaatoms are added to the surface. In this case, the N^2 SNR growth from (9.32) dominates. By contrast, when the static path is strong, the capacity is already quite high for $N = 0$, and a huge surface is needed before it has a noticeable impact on the capacity. The relative strength of the propagation path provided by the surface matters, not how much power it provides in an absolute sense. This indicates that reconfigurable surfaces are particularly valuable in deployment scenarios with weak static paths, where even a small surface makes a significant difference.

While Section 9.1 considered reflections in LOS scenarios, we have not assumed any specific channel model in this section. By optimizing the reflection matrix, we can find the phase-shift profile of a surface with fixed dimensions that maximizes the received signal power in a given propagation environment. We could mechanically bend and deform a homogeneous surface to obtain the corresponding physical shape, but instead, a reconfigurable surface synthesizes that shape using a heterogeneous impedance pattern. This

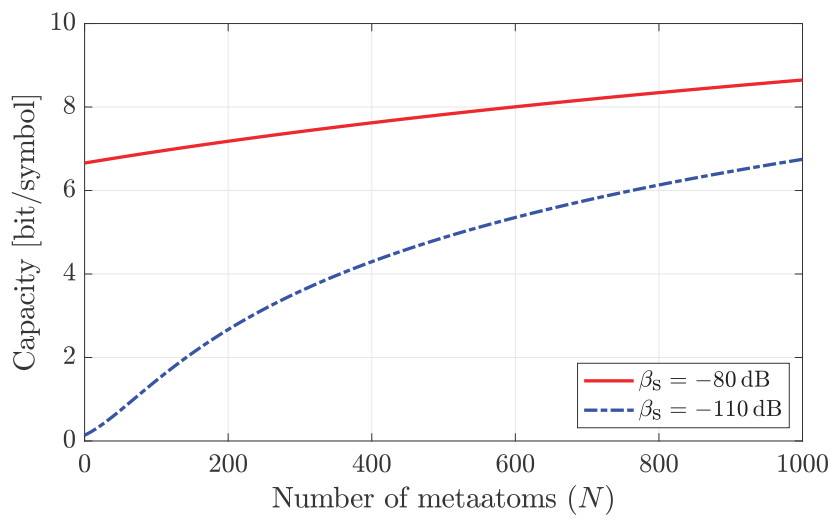


Figure 9.11: The capacity of a SISO channel that is aided by a reconfigurable surface. The capacity increases with the number of metaatoms, and the relative improvement is particularly large when the channel gain β_s of the static path is weak.

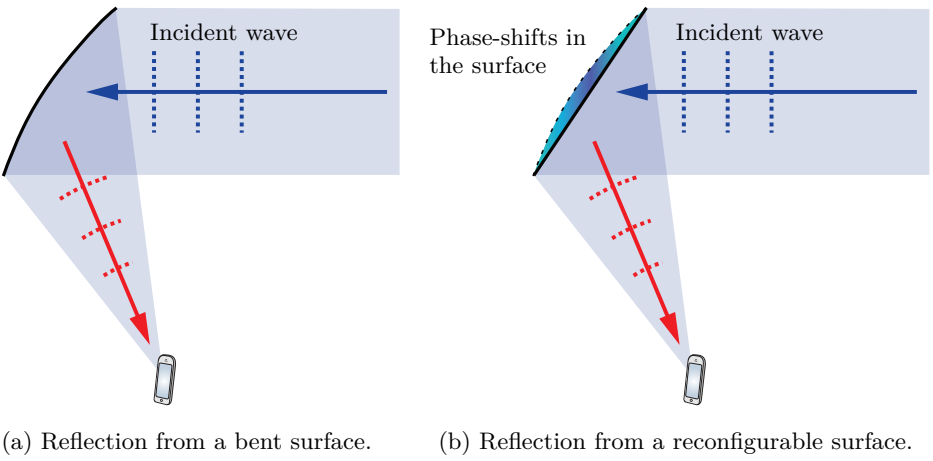


Figure 9.12: The shape and electric properties of the reflecting surface jointly determine the reflection direction. In (a), a homogenous surface is bent to reflect the incident wave toward the receiver. In (b), a flat reconfigurable surface has a phase-shift profile that achieves the same result by adding extra phase-shifts to signals in the center compared to the edges.

enables us to reconfigure the reflection properties rapidly when the environment or transmitter/receiver locations change. Figure 9.12 illustrates this principle. When the wave reaches the surface from the right, a parabolically bent surface will reflect the signal toward the indicated receiver, as shown in Figure 9.12(a). A flat, reconfigurable surface can synthesize the same reflection behavior by adding extra phase-shifts to the wave components reflected at

its center compared to its edges. This is illustrated in Figure 9.12(b), where the coloring behind the surface represents the phase-shifts using the same scale as in Figure 9.7. The phase-shifts are negative in this scenario since they represent extra delays incurred by the metaatoms to synthesize a bent surface.

9.2.1 Line-of-Sight Channel Modeling and Surface Placement

The channel gain can be computed precisely in free-space LOS propagation using the formulas provided in Section 1.1.4. Suppose the distance from the transmitter to the surface is d_t and the distance from the surface to the receiver is d_r . If the transmitter has the antenna gain $G_t(\varphi_t, \theta_t)$ towards the surface and each small metaatom has the area A_m (i.e., the antenna gain is $\frac{4\pi}{\lambda^2} A_m$), it follows from (1.40) that the channel gain between them is

$$\beta_t = \frac{\lambda^2}{(4\pi d_t)^2} G_t(\varphi_t, \theta_t) \frac{4\pi}{\lambda^2} A_m = \frac{G_t(\varphi_t, \theta_t) A_m}{4\pi d_t^2}. \quad (9.33)$$

Similarly, if the receiver has the antenna gain $G_r(\varphi_r, \theta_r)$ towards the surface, then the channel gain between them is

$$\beta_r = \frac{\lambda^2}{(4\pi d_r)^2} G_r(\varphi_r, \theta_r) \frac{4\pi}{\lambda^2} A_m = \frac{G_r(\varphi_r, \theta_r) A_m}{4\pi d_r^2}. \quad (9.34)$$

When the propagation path via the surface dominates over the static path, it follows from (9.32) that the end-to-end channel gain can be expressed as

$$N^2 \beta_r \beta_t = N^2 \frac{G_t(\varphi_t, \theta_t) A_m}{4\pi d_t^2} \frac{G_r(\varphi_r, \theta_r) A_m}{4\pi d_r^2} = \frac{N^2 G_t(\varphi_t, \theta_t) G_r(\varphi_r, \theta_r) A_m^2}{(4\pi d_t d_r)^2}, \quad (9.35)$$

where we utilized the gain expressions in (9.33) and (9.34). There are many squares in (9.35) because the end-to-end channel gain is the product of two conventional channel gain expressions. First, the metaatom's area is squared because it appears in both gain expressions. It is also multiplied by N^2 , which implies that it is the total area NA_m of the surface that determines the end-to-end gain. Second, the squared propagation distances d_t^2 and d_r^2 appear in the expression since the signal power attenuates inversely proportional to them in free-space propagation. The distances are also multiplied together.

In NLOS propagation scenarios, any channel model could be used for the individual channels. The only important aspect is to account for the small area A_m of each metaatom, which results in a gain $\frac{4\pi}{\lambda^2} A_m$. This value is smaller than one since we consider sufficiently small-sized metaatoms to be able to capture and retransmit power almost isotropically.

It is not only the number of metaatoms that determines the end-to-end channel gain but also where the reconfigurable surface is deployed. Ideally, the transmitter and receiver should have LOS channels to the surface because these are generally stronger than NLOS channels. When multiple deployment locations satisfy that condition, further characteristics can be considered. The following example highlights one key property.

Example 9.5. Suppose a transmitter and a receiver with isotropic antennas are located in the same horizontal plane. They are both 10 m from a long wall but 100 m apart from each other. There is no static path between them, but they can see any point on the wall. Where on the wall should a reconfigurable surface be deployed to maximize the end-to-end channel gain?

We let N denote the number of metaatoms, A_m denote the area per metaatom, and $\sqrt{10^2 + d_w^2}$ be the distance between the transmitter and the surface, where $d_w \in [0, 100]$ m is the distance along the wall. The distance between the surface and the receiver is then given as $\sqrt{10^2 + (100 - d_w)^2}$. Since we have LOS channels and isotropic transmit and receive antennas, the end-to-end channel gain can be expressed using (9.35) as

$$N^2 \beta_r \beta_t = N^2 \frac{A_m}{4\pi(10^2 + (100 - d_w)^2)} \frac{A_m}{4\pi(10^2 + d_w^2)}. \quad (9.36)$$

The deployment location affects the term $(10^2 + (100 - d_w)^2)(10^2 + d_w^2)$ in the denominator. This term has the first-order derivative

$$\begin{aligned} \frac{\partial}{\partial d_w} (10^2 + (100 - d_w)^2) (10^2 + d_w^2) &= 4d_w^3 - 600d_w^2 + 20400d_w - 20000 \\ &= 4(d_w - 50) (d_w^2 - 100d_w + 100), \end{aligned} \quad (9.37)$$

which has the roots $d_w = 50$ m, $d_w = 50 - 20\sqrt{6} \approx 1$ m, and $d_w = 50 + 20\sqrt{6} \approx 99$ m. The former value is a maximum and the latter values are two minima, as can be proved by checking the signs of the second-order derivative. Hence, the channel gain is minimized when the surface is deployed in the middle and maximized when it is close to the transmitter or receiver.

The conclusion from this example is that we should look for all deployment locations where the surface has LOS conditions to both the base station and prospective user locations. Among these locations, we should pick the one that is closest to either of them since this maximizes the channel gain. This insight motivates the holographic MIMO architecture, mentioned briefly in Section 7.4.2, where a metasurface is deployed as a part of the base station to create an analog beamforming architecture with small antenna spacing. In this chapter, the reconfigurable surface is meant to be decoupled from the transmitter and receiver, but it should preferably be quite near one of them.

Figure 9.13 shows the end-to-end channel gain in (9.36) for $N = 200$ metaatoms that each has the area $A_m = (\lambda/4)^2$ where $\lambda = 0.1$ m is the wavelength. As expected from the example above, the maximum channel gain is achieved close to either the transmitter or receiver, while the minimum value is obtained in the middle. The difference is around 8 dB in this example, which is substantial but not huge compared to the fact that all the considered channel gains are at the order of -100 dB.

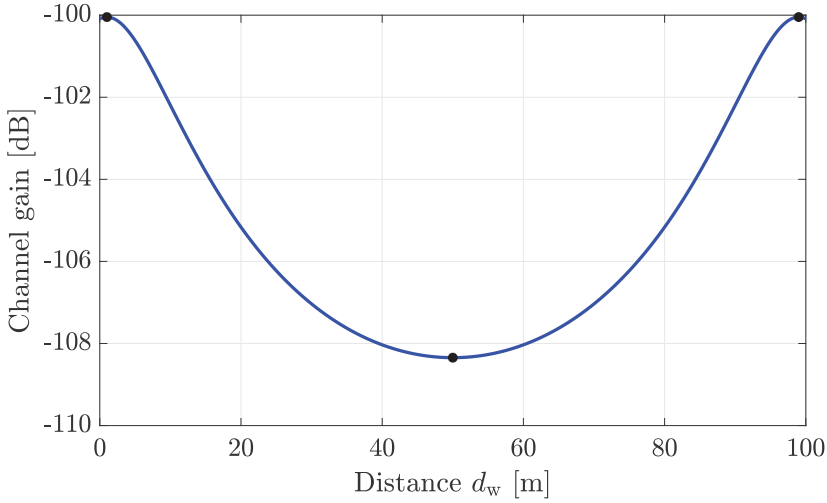


Figure 9.13: The end-to-end channel gain in (9.36) depending on how far the reconfigurable surface is from the transmitter, when the total distance between the transmitter and receiver is 100 m. The stars show the maximum and minimum values, which were derived in Example 9.5.

9.2.2 Acquiring Channel State Information and Feedback Signaling

The reconfigurable surface requires CSI to compute the capacity-maximizing reflection matrix. Specifically, (9.27) shows that it must know the phase of the static channel $\arg(h_s)$ and the phases $\arg(h_{r,n}h_{t,n})$ of the paths through each of the N metaatoms. It is sufficient to know the phase of the *cascaded channel coefficient* $h_{r,n}h_{t,n}$, while the individual characteristics of $h_{r,n}$ and $h_{t,n}$ are unimportant. These $N + 1$ real-valued phase coefficients can be estimated by sending a pilot sequence similar to the point-to-point scenario described in Section 4.2.4. The precise details are somewhat different since the reconfigurable surface can only reflect signals, not measure them. To describe the procedure, we begin by factorizing the end-to-end channel in (9.22) as

$$h = h_s + \sum_{n=1}^N h_{r,n} e^{j\psi_n} h_{t,n} = \underbrace{\begin{bmatrix} 1 \\ e^{j\psi_1} \\ \vdots \\ e^{j\psi_N} \end{bmatrix}^T}_{=\psi^T} \underbrace{\begin{bmatrix} h_s \\ h_{r,1}h_{t,1} \\ \vdots \\ h_{r,N}h_{t,N} \end{bmatrix}}_{=\tilde{\mathbf{h}}}, \quad (9.38)$$

where $\psi \in \mathbb{C}^{N+1}$ contains all the configurable phase-shifts (including a 1 for the static path) and $\tilde{\mathbf{h}} \in \mathbb{C}^{N+1}$ contains all the necessary channel coefficients. This vector was previously defined in (9.30). Every time a signal is transmitted over the channel, it will experience the scalar channel coefficient h obtained as the inner product between the channel vector $\tilde{\mathbf{h}}$ and the complex conjugate of the phase-shift vector ψ . Hence, only the one-dimensional part of the

$(N+1)$ -dimensional channel vector that is aligned with $\boldsymbol{\psi}^*$ is utilized to reflect the signal toward the receiver, while the remaining dimensions are invisible to the receiver. For the receiver to observe the entire channel vector, we must send multiple pilot signals and vary the phase-shift configuration vector to explore all the dimensions of $\mathbb{C}^{N+1}$ where $\check{\mathbf{h}}$ might have a component.

By following the notation from Section 4.2.4, we consider the transmission of a preamble of length L_p designed to enable channel estimation. Specifically, a constant pilot sequence $x[l] = \sqrt{q}$ is transmitted for $l = 1, \dots, L_p$ and we reflect it using a sequence of different configuration vectors: $\boldsymbol{\psi}[1], \dots, \boldsymbol{\psi}[L_p] \in \mathbb{C}^{N+1}$. The received signal at time instance l can then be expressed as

$$y[l] = \boldsymbol{\psi}^T[l] \check{\mathbf{h}} \sqrt{q} + n[l], \quad l = 1, \dots, L_p, \quad (9.39)$$

but it is convenient to write it in matrix/vector form as

$$\underbrace{\begin{bmatrix} y[1] \\ \vdots \\ y[L_p] \end{bmatrix}}_{=\check{\mathbf{y}}} = \underbrace{\begin{bmatrix} \boldsymbol{\psi}[1] & \dots & \boldsymbol{\psi}[L_p] \end{bmatrix}^T}_{=\boldsymbol{\Psi}} \check{\mathbf{h}} \sqrt{q} + \underbrace{\begin{bmatrix} n[1] \\ \vdots \\ n[L_p] \end{bmatrix}}_{=\mathbf{n}}. \quad (9.40)$$

If the channel vector $\check{\mathbf{h}}$ is treated as deterministic but unknown, the PDF of $\check{\mathbf{y}} \in \mathbb{C}^{L_p}$ can be expressed using (2.80) as

$$f_{\check{\mathbf{y}}}(\check{\mathbf{y}}) = \frac{1}{(\pi N_0)^{L_p}} e^{-\frac{\|\check{\mathbf{y}} - \boldsymbol{\Psi} \check{\mathbf{h}} \sqrt{q}\|^2}{N_0}} \quad (9.41)$$

because $\check{\mathbf{y}} - \boldsymbol{\Psi} \check{\mathbf{h}} \sqrt{q} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, N_0 \mathbf{I}_{L_p})$. The ML estimate $\hat{\mathbf{h}}$ of $\check{\mathbf{h}}$ is the vector that maximizes the PDF, which corresponds to minimizing the squared norm expression in its exponent. By equating the argument of the norm to zero, we obtain

$$\check{\mathbf{y}} - \boldsymbol{\Psi} \hat{\mathbf{h}} \sqrt{q} = \mathbf{0} \quad \Rightarrow \quad \hat{\mathbf{h}} = \frac{1}{\sqrt{q}} \boldsymbol{\Psi}^{-1} \check{\mathbf{y}} \quad (9.42)$$

if the matrix $\boldsymbol{\Psi} \in \mathbb{C}^{L_p \times (N+1)}$ is invertible. This requires that $L_p = N+1$ since only square matrices are invertible. Moreover, we need to find an invertible matrix of that size that satisfies two conditions: all entries of the first column are equal to 1, and all other entries are complex exponentials that can be implemented using the phase-shifting ability of a reconfigurable surface. Both properties are satisfied by the DFT matrix in (2.198) if it is scaled properly:

$$\boldsymbol{\Psi} = \sqrt{N+1} \mathbf{F}_{N+1}. \quad (9.43)$$

For this particular choice, the ML estimate in (9.42) can be expressed as

$$\hat{\mathbf{h}} = \frac{1}{\sqrt{q}} \boldsymbol{\Psi}^{-1} \left(\boldsymbol{\Psi} \check{\mathbf{h}} \sqrt{q} + \mathbf{n} \right) = \check{\mathbf{h}} + \frac{1}{\sqrt{q(N+1)}} \mathbf{F}_{N+1}^H \mathbf{n}, \quad (9.44)$$

which is the true channel vector plus a scaled noise term with i.i.d. entries distributed as $\mathcal{N}_{\mathbb{C}}(0, N_0/(q(N+1)))$. The estimation error vanishes as $q \rightarrow \infty$ as expected from a well-crafted estimator.

Example 9.6. Can an ML estimate of $\check{\mathbf{h}}$ be computed if $L_p \neq N + 1$?

Yes, we can always do our best to maximize the PDF in (9.41) even if the performance varies. If $L_p > N + 1$, we can pick Ψ as a full-rank matrix. We still want to solve the linear system of equations $\check{\mathbf{y}} - \Psi\check{\mathbf{h}}\sqrt{q} = \mathbf{0}$ with respect to $\check{\mathbf{h}}$. This system is overdetermined and might lack a solution. However, the channel vector can only be observed in the subspace spanned by the columns of Ψ ; thus, we can project the equation to that subspace and then solve it:

$$\Psi^H \check{\mathbf{y}} - \Psi^H \Psi \hat{\mathbf{h}} \sqrt{q} = \mathbf{0} \Rightarrow \hat{\mathbf{h}} = \frac{1}{\sqrt{q}} (\Psi^H \Psi)^{-1} \Psi^H \check{\mathbf{y}} = \check{\mathbf{h}} + \frac{1}{\sqrt{q}} (\Psi^H \Psi)^{-1} \Psi^H \mathbf{n}. \quad (9.45)$$

The matrix $(\Psi^H \Psi)^{-1} \Psi^H$ is called the left pseudo-inverse of Ψ . The estimate is more precise than with $L_p = N + 1$ since it builds on more observations.

If $L_p < N + 1$, the linear system of equations $\check{\mathbf{y}} - \Psi\check{\mathbf{h}}\sqrt{q} = \mathbf{0}$ is underdetermined and has many solutions. Instead of picking an arbitrary solution, which leads to estimation errors that remain as $q \rightarrow \infty$, it can be desirable to reformulate the entire problem to reduce the number of unknowns. We can group N_s adjacent metaatoms together into a subarray that must use the same phase-shift value, motivated by the fact that the optimal phase-shift pattern often varies slowly over the surface. A single channel coefficient can then represent the cascaded channel through one subarray. Hence, in the reformulated estimation problem, detailed in [160], there are only $N/N_s + 1$ unknown coefficients. For any $L_p \geq 2$, we can pick the subarray size N_s such that $L_p \geq N/N_s + 1$ to avoid an underdetermined estimation problem.

The ML estimate is computed at the receiver, not the reconfigurable surface that needs it. A possible solution is that the receiver (e.g., the base station) computes the estimate, then determines the desirable configuration by putting the estimates into (9.27), and finally feeds this information back to the surface. This procedure is illustrated in Figure 9.14. The feedback link requires the reconfigurable surface to be equipped with a transceiver.

We will now compare the capacity-maximizing configuration (based on perfect CSI) with the capacities obtained when the reconfigurable surface is tuned based on the ML estimate (imperfect CSI) and when random phase-shifts uniformly distributed in $[-\pi, \pi)$ are used. Since randomness affects h in the latter cases, we present the average capacity values in Figure 9.15. These values are computed assuming the receiver knows h perfectly during data transmission, so the randomness only affects how the surface is configured. We consider $N = 200$ metaatoms, $\beta_t = -80$ dB, $\beta_r = -60$ dB, and $\beta_s = -110$ dB (as in Figure 9.11). The SNR shown on the horizontal axis is defined based on the static path as $q\beta_s/N_0$. The estimation accuracy is low when the SNR is small; thus, random phase-shifts give the same capacity as when the ML

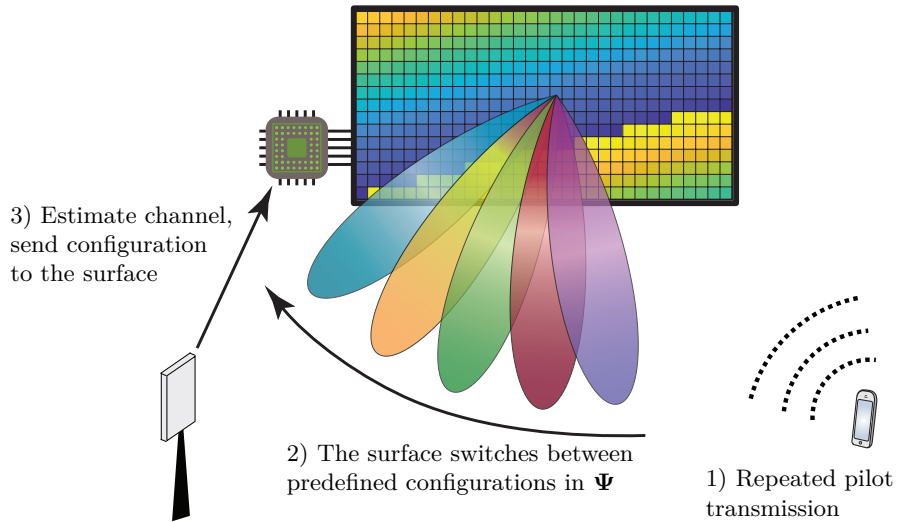


Figure 9.14: A reconfigurable surface can be configured by letting the transmitter repeat a pilot transmission while the surface reflects it using different predefined configurations. The receiver then computes an estimate of the channel vector and uses it to compute the desirable configuration. This information is then sent to the surface using a feedback link.

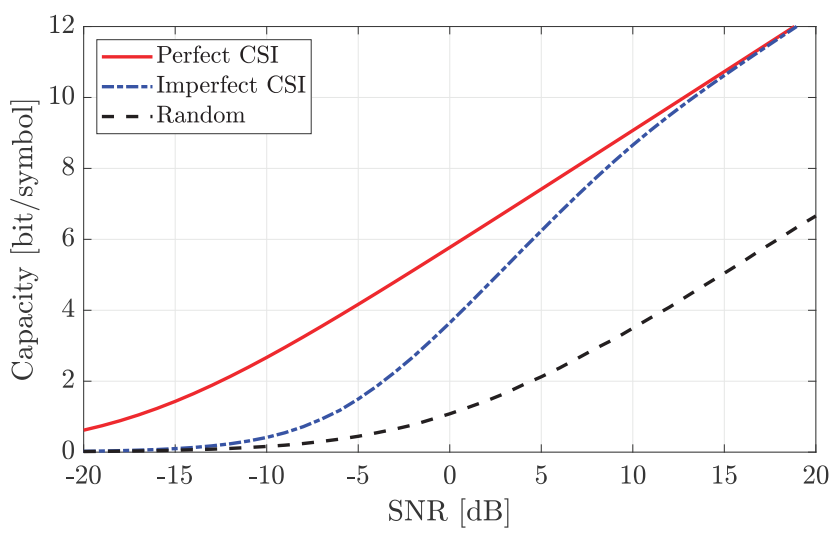


Figure 9.15: The capacity as a function of the SNR, considering a SISO channel aided by a reconfigurable surface with $N = 200$ metaatoms. Three different configurations are compared: the capacity-maximizing one based on perfect CSI, one based on the ML estimator (imperfect CSI), and one using random phase-shifts.

estimate is used for selecting the phases. As the SNR increases, the imperfect CSI curve improves rapidly, thanks to the better estimation accuracy, and converges to the perfect CSI curve. The gap to the case with random phase-shifts is then large. It is important to note that the SNR per metaatom, $q\beta_r\beta_t/N_0$, is 30 dB smaller than what is shown on the horizontal axis; on the other hand, the 201-length pilot sequence increases the SNR during the channel estimation by $10\log_{10}(201) \approx 23$ dB. In conclusion, assessing what SNR value is small versus large in this context is complicated.

The ML estimator derived and discussed above is non-parametric, which means that we look for any conceivable channel vector in $\mathbb{C}^{N+1}$. When the ML estimation framework was previously applied in Section 4.2.5, we restricted the search to LOS channels that are parametrized by the angle-of-arrival to a multi-antenna receiver. A similar parametric ML estimator can be developed when the channels to and from the surface are array response vectors, but we refer to [161] for the precise details.

9.3 Wideband Communication using Reconfigurable Surfaces

In this section, we will analyze how reconfigurable surfaces can be utilized to enhance communication over wideband channels. To ensure we capture the essential new characteristics, we must revisit how practical continuous passband channels were transformed into discrete complex baseband channels in Section 2.3. The previous analysis considered the general setup in Figure 9.16(a), where a passband signal $z_p(t)$ is transmitted over a wireless channel and $v_p(t)$ denotes the filtered version that reaches the receiver before noise is added to it. The channel was described by the impulse response $g_p(t)$, and it depends on the propagation environment that the reconfigurable surface can control. Hence, we will now denote the impulse response as $g_{p;\psi}(t)$, where the vector ψ represents the surface configuration.

The end-to-end channel impulse response $g_{p;\psi}(t)$ is the sum of the impulse responses of the different propagation paths. We begin by defining the impulse response $g_{s,p}(t)$ of the static LTI channel that the signal propagates over in the absence of the reconfigurable surface. The transmitted signal $z_p(t)$ also propagates to each of the N metaatoms in the reconfigurable surface through a separate LTI channel represented by an arbitrary impulse response $g_{t,n,p}(t)$, for $n = 1, \dots, N$. When the signal reaches metaatom n , it will be filtered by its internal circuitry and then reradiated. The filtering happens in the analog domain and will be modeled as an LTI filter. We denote the impulse response as $\vartheta_{n,p;\psi_n}(t)$ and stress that it is reconfigurable in the sense that it depends on an external stimulus represented by the variable ψ_n from the vector ψ . In other words, we can choose from a set of possible impulse responses by selecting ψ_n . To be consistent with the LTI assumption, only one value of ψ_n can be used during the considered signal transmission.

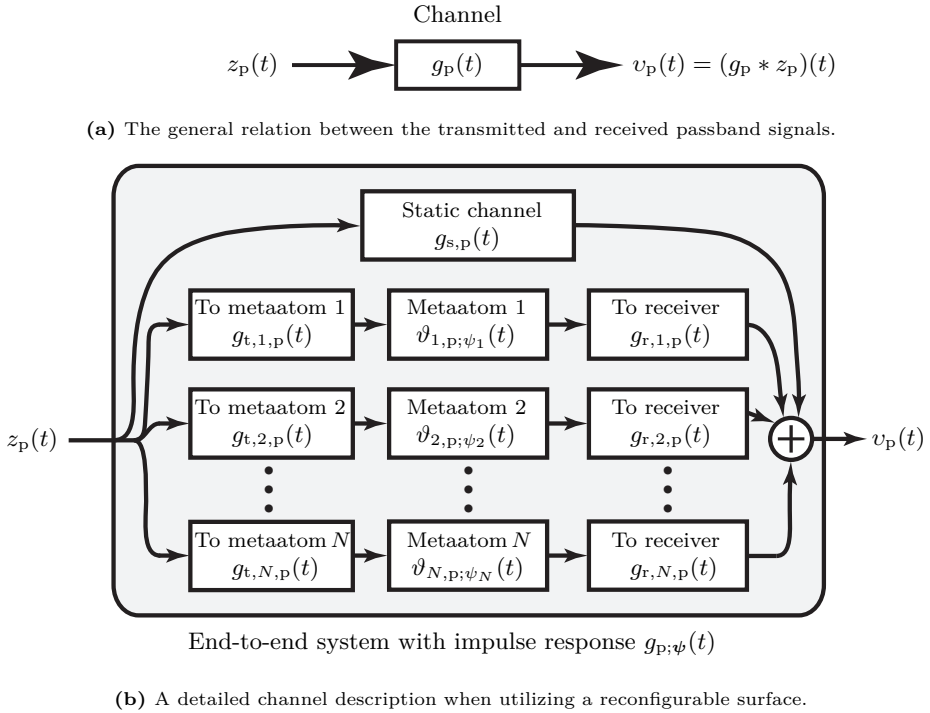


Figure 9.16: The received passband signal $v_p(t)$ is the convolution between the transmitted signal $z_p(t)$ and the channel impulse response $g_p(t)$, as shown in (a). When utilizing a reconfigurable surface with N metaatoms, the impulse response is the superposition/addition of the impulse response $g_{s,p}(t)$ of the static channel and the N controllable impulse responses of the channels via each of the N metaatoms, as shown in (b).

When the signal is reradiated from metaatom n , it propagates to the receiver over yet another LTI channel with an arbitrary impulse response $g_{r,n,p}(t)$. Since the transmitted signal propagates via metaatom n to the receiver through a cascade of three LTI filters, the joint impulse response is the convolution of their impulse responses: $(g_{r,n,p} * \vartheta_{n,p;\psi_n} * g_{t,n,p})(t)$. We thereby obtain the input-output relation illustrated in Figure 9.16(b):

$$\begin{aligned}
 v_p(t) &= (g_{s,p} * z_p)(t) + \sum_{n=1}^N (g_{r,n,p} * \vartheta_{n,p;\psi_n} * g_{t,n,p} * z_p)(t) \\
 &= \left(\underbrace{\left[g_{s,p} + \sum_{n=1}^N g_{r,n,p} * \vartheta_{n,p;\psi_n} * g_{t,n,p} \right]}_{=g_{p;\psi}} * z_p \right)(t), \quad (9.46)
 \end{aligned}$$

where we identify the impulse response of the end-to-end system as

$$g_{p;\psi}(t) = g_{s,p}(t) + \sum_{n=1}^N (g_{r,n,p} * \vartheta_{n,p;\psi_n} * g_{t,n,p})(t). \quad (9.47)$$

We recall from (2.116)-(2.117) that filtering of a passband signal can be represented by filtering the equivalent complex-baseband signal using impulse responses that are downshifted. By applying this principle to each component of $g_{p;\psi}(t)$, the complex baseband representation of (9.46) becomes

$$v(t) = (g_s * z)(t) + \sum_{n=1}^N (g_{r,n} * \vartheta_{n;\psi_n} * g_{t,n} * z)(t), \quad (9.48)$$

where the impulse responses of the downshifted channels and filters are defined as

$$g_s(t) = g_{s,p}(t)e^{-j2\pi f_c t}, \quad (9.49)$$

$$g_{t,n}(t) = g_{t,n,p}(t)e^{-j2\pi f_c t}, \quad (9.50)$$

$$g_{r,n}(t) = g_{r,n,p}(t)e^{-j2\pi f_c t}, \quad (9.51)$$

$$\vartheta_{n;\psi_n}(t) = \vartheta_{n,p;\psi_n}(t)e^{-j2\pi f_c t}. \quad (9.52)$$

The end-to-end channel has the impulse response

$$g_\psi(t) = g_s(t) + \sum_{n=1}^N (g_{r,n} * \vartheta_{n;\psi_n} * g_{t,n})(t). \quad (9.53)$$

We notice that the convolution of a chain of impulse responses in the passband becomes the convolution of the corresponding chain of complex-baseband impulse responses. This property seems natural but is actually a feature of the definitions previously made in Section 2.3.1. We considered a passband signal that is sent over a channel with arbitrary frequency support and defined how the signal and channel/filter are transformed to the baseband differently. This can be called the *pseudo-baseband representation* since the channel is not a baseband filter, but the output is a baseband signal since the input is a baseband signal. By contrast, many other textbooks consider a stricter complex-baseband representation, where each channel is a passband filter that is transformed to the baseband identically to the signal. That definition is less practical since wireless channels are not passband filters but support signals of any frequency. More importantly, it gives rise to extra scaling factors, and these multiply when considering convolutions of filters, making the stricter model inappropriate when studying reconfigurable surfaces.

When a discrete sequence of data symbols $x[k]$ is transmitted using PAM, bandpass filtered at the receiver, and sampled on the symbol rate, the resulting received signal sequence $y[l]$ was expressed in (7.7) as

$$y[l] = \sum_{\ell=0}^T h_\psi[\ell]x[l-\ell] + n[l], \quad (9.54)$$

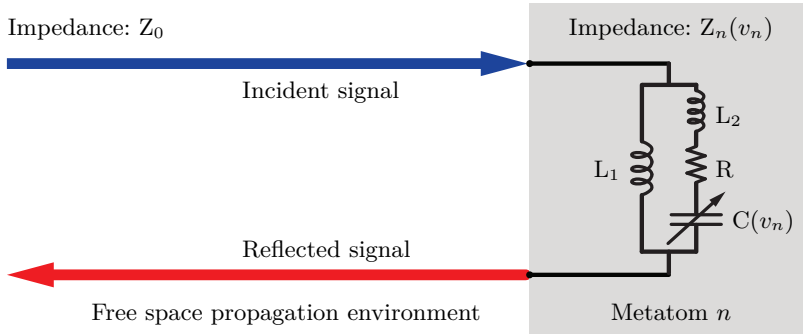


Figure 9.17: An example of a lumped-element model of a metaatom containing two inductors (L_1, L_2), one resistor (R), and a varactor with a controllable capacitance $C(v_n)$.

where the $T + 1$ discrete-time channel coefficients $h_\psi[0], \dots, h_\psi[T]$ are computed based on the end-to-end channel model in (9.53) as

$$\begin{aligned} h_\psi[\ell] &= (p * g_\psi * p)(t) \Big|_{t=\ell/B} \\ &= (p * g_s * p)(t) \Big|_{t=\ell/B} + \sum_{n=1}^N (p * g_{r,n} * \vartheta_{n;\psi_n} * g_{t,n} * p)(t) \Big|_{t=\ell/B}. \end{aligned} \quad (9.55)$$

Conventional propagation models can be used for the impulse responses $g_s(t)$, $g_{t,n}(t)$, $g_{r,n}(t)$ of the wireless channels, by accounting for the effective areas of antennas and metaatoms. To compute an expression of (9.55), we must also characterize the impulse response $\vartheta_{n;\psi_n}(t)$ of a metaatom.

9.3.1 Impulse Response of a Metaatom

We will showcase a basic model of the impulse response $\vartheta_{n;\psi_n}(t)$ of a metaatom by analyzing a practical implementation. Figure 9.17 shows a lumped-element model of metaatom n containing two parallel branches where the first contains an inductor with inductance L_1 and the second contains a series with an inductor with inductance L_2 , a resistor with resistance R , and a varactor with a capacitance $C(v_n)$ controlled by the bias voltage v_n . This parallel resonance circuit is a simplified version of the metaatom design in [162] and was considered for reconfigurable surfaces in [163]. Using circuit theory methods, the impedance of the metaatom can be shown to be

$$Z_n(v_n) = \frac{j2\pi f L_1 \left(j2\pi f L_2 + R + \frac{1}{j2\pi f C(v_n)} \right)}{j2\pi f L_1 + \left(j2\pi f L_2 + R + \frac{1}{j2\pi f C(v_n)} \right)}, \quad (9.56)$$

for a signal with the frequency f . We can compute the frequency-dependent reflection coefficient by substituting this expression into (9.13).

Figure 9.18 shows the frequency response of the metaatom for frequencies around an intended carrier frequency of $f_c = 3$ GHz. The parameters

of the lumped-element model in Figure 9.17 are $L_1 = 2.5 \text{ nH}$, $L_2 = 0.7 \text{ nH}$, $Z_0 = 377 \text{ ohm}$, and $R = 1 \text{ ohm}$. Since the reflection coefficient is complex, the phase and amplitude responses are presented in (a) and (b), respectively. There are curves in Figure 9.18(a) for four different capacitance values obtained by controlling the bias voltage of the varactor. The specific values have been selected to give the phase-shifts $3\pi/4, \pi/4, -\pi/4, -3\pi/4$ at the carrier frequency. The phase begins close to $+\pi$ on all the curves because the reradiated electric field is inverted (upside down). The phase variations are large when considering the GHz range, which is natural for all filters. The simplest representation of reflection would be a pure time delay τ , which has the impulse response $\delta(t - \tau)$ and frequency response $e^{-j2\pi f\tau}$ with a phase that varies linearly with the frequency. Since the curves are approximately linear next to the carrier frequency, this model is suitable if the signal bandwidth B is limited to a few hundred MHz. Outside this range, the phase response curves have non-linear shapes that will distort the signal in the time domain; thus, a metaatom has a limited useful bandwidth range. As long as the system uses a smaller bandwidth, the reflected signal will be undistorted, and the propagation channels will determine whether the communication system is narrowband or wideband. The four curves have roughly the same shape but are shifted in the frequency domain; it is this shift that the varactor controls.

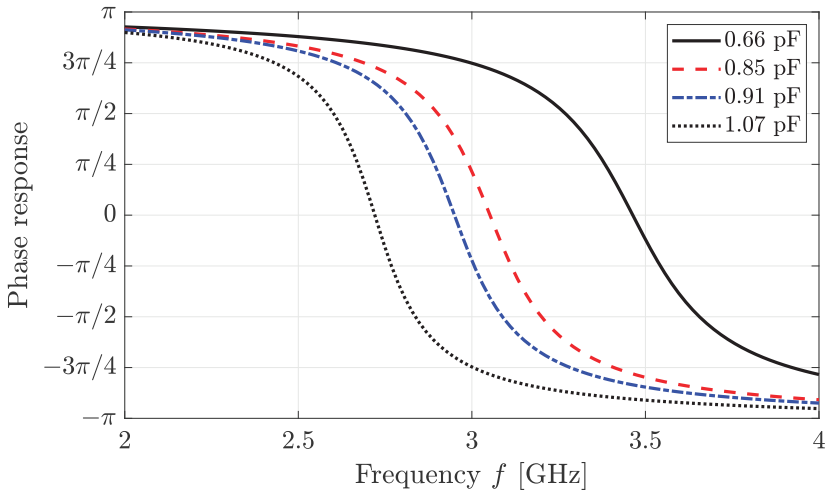
The amplitude responses are shown in Figure 9.18(b) for the same capacitance values, but different resistances: $R = 1 \text{ ohm}$ and $R = 0 \text{ ohm}$. The theoretical maximum amplitude response is 0 dB since the metaatom is a passive circuit that reflects the signal without amplification.⁴ All the signal power is reflected when the resistance is negligible, while there are a few dB of amplitude losses when the resistance is non-zero. In the latter case, the amplitude loss is also frequency-dependent. The loss is largest at the frequency where the phase response is zero due to resonance in the circuit. While building metaatoms with minimal reflection losses is desirable, we should keep in mind that a few dB is minor compared to the propagation losses over wireless channels that are typically at the order of 100 dB.

In summary, an ideal metaatom design has no amplitude losses and a linear phase within the signal band. Hence, its impulse response can be expressed as $\vartheta_{n,p;\psi_n}(t) = \delta(t - \tau_{\psi_n})$ in the passband, which results in the downshifted filter

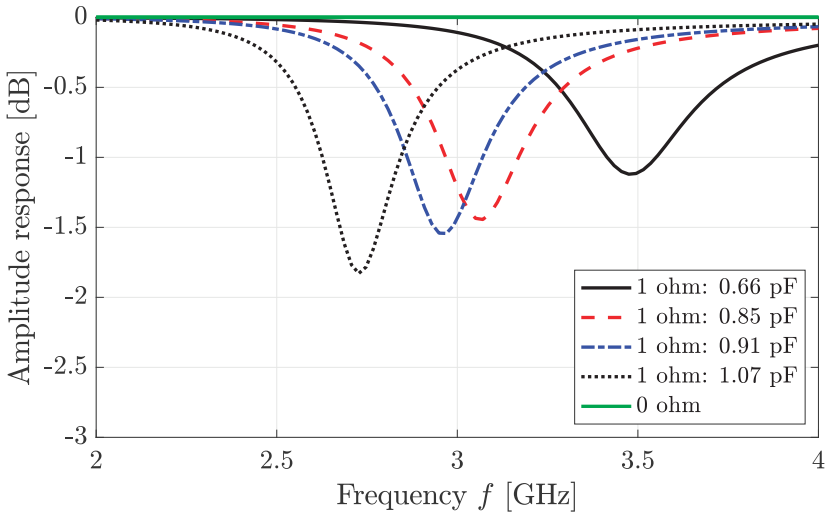
$$\vartheta_{n;\psi_n}(t) = \delta(t - \tau_{\psi_n})e^{-j2\pi f_c t}, \quad (9.57)$$

where the controllable delay is denoted by τ_{ψ_n} . We will soon show that (9.57) results in a phase-shift of $\psi_n = -2\pi f_c \tau_{\psi_n}$ in the system model; thus, the phase-shift caused by a metaatom is controlled by tuning the reflection delay.

⁴The passiveness is a key feature since active components (e.g., an amplifier) add noise to the reflected signal, which is not the case when using reconfigurable surfaces. Retransmitting devices that contain amplifiers are normally referred to as repeaters or relays and must be studied using different system models than in this chapter.



(a) Phase response for different frequencies when $R = 1$ ohm.



(b) Amplitude response for different frequencies.

Figure 9.18: The frequency response when a metaatom with the impedance in (9.56) reflects a signal in free space. The curves are obtained for different capacitances of the varactor, which are selected to give the phase-shifts $3\pi/4$, $\pi/4$, $-\pi/4$, $-3\pi/4$ at 3 GHz. The phase response is shown in (a) and the the amplitude response is shown in (b).

9.3.2 OFDM-Based Channel Model with a Reconfigurable Surface

In this section, we will determine the channel coefficients in an OFDM system aided by a reconfigurable surface. We can use the basic channel model from (2.124) to express a static channel with L_s propagation paths as

$$g_{s,p}(t) = \sum_{i=1}^{L_s} \alpha_{s,i} \delta(t + \eta - \tau_{s,i}) \quad \Rightarrow \quad g_s(t) = \sum_{i=1}^{L_s} \alpha_{s,i} e^{-j2\pi f_c t} \delta(t + \eta - \tau_{s,i}), \quad (9.58)$$

where $\alpha_{s,i} \in [0, 1]$ is the attenuation and $\tau_{s,i} \geq 0$ is the delay of path i , for $i = 1, \dots, L_s$. We recall that η denotes the receiver's clock delay, which ensures that the receiver takes samples when the signal reaches it and not when the signal leaves the transmitter. This parameter must be selected as described in Section 7.1 to achieve the causal FIR filter representation in (9.54). We denote the number of propagation paths between the transmitter and reconfigurable surface as L_t and between the surface and receiver as L_r . Similarly to (9.58), we can then model the impulse responses to and from metaatom n as $g_{t,n,p}(t) = \sum_{i=1}^{L_t} \alpha_{t,n,i} \delta(t - \tau_{t,n,i})$ and $g_{r,n,p}(t) = \sum_{j=1}^{L_r} \alpha_{r,n,j} \delta(t + \eta - \tau_{r,n,j})$. These can be expressed in the complex baseband as

$$g_{t,n}(t) = \sum_{i=1}^{L_t} \alpha_{t,n,i} e^{-j2\pi f_c t} \delta(t - \tau_{t,n,i}), \quad (9.59)$$

$$g_{r,n}(t) = \sum_{j=1}^{L_r} \alpha_{r,n,j} e^{-j2\pi f_c t} \delta(t + \eta - \tau_{r,n,j}), \quad (9.60)$$

where $\alpha_{t,n,i}, \alpha_{r,n,j} \in [0, 1]$ are the attenuations and $\tau_{t,n,i}, \tau_{r,n,j} \geq 0$ are the propagation delays. Note that the receiver's timing delay η is only included in the channels that lead to the receiver. By substituting the metaatom's impulse response in (9.57) and the channel impulse responses in (9.58)–(9.60) into (9.55), the channel coefficients can be computed as⁵

$$h_{\psi}[\ell] = \sum_{i=1}^{L_s} \alpha_{s,i} e^{-j2\pi f_c (\tau_{s,i} - \eta)} \text{sinc}(\ell + B(\eta - \tau_{s,i})) + \sum_{n=1}^N \sum_{j=1}^{L_r} \sum_{i=1}^{L_t} \alpha_{r,n,j} \alpha_{t,n,i} e^{-j2\pi f_c (\tau_{r,n,j} + \tau_{t,n,i} + \tau_{\psi_n} - \eta)} \text{sinc}(\ell + B(\eta - \tau_{r,n,j} - \tau_{t,n,i})) \quad (9.61)$$

for $\ell = 0, \dots, T$, by utilizing the facts that $(p * p)(t) = \text{sinc}(Bt)$ and that the convolution between $\text{sinc}(Bt)$ and $e^{-j2\pi f_c t} \delta(t - \tau)$ is $\text{sinc}(B(t - \tau)) e^{-j2\pi f_c \tau}$.

⁵The last sinc-term in (9.61) becomes $\text{sinc}(\ell + B(\eta - \tau_{r,n,j} - \tau_{t,n,i} - \tau_{\psi_n}))$ but the term containing τ_{ψ_n} can be dropped since the delay caused by the reflection is negligible compared to the symbol time $1/B$, in the sense that $\tau_{\psi_n}/(1/B) = B\tau_{\psi_n} \approx 0$. The metaatom nevertheless creates a noticeable phase-shift since $f_c \gg B$.

We notice that (9.61) contains L_s static paths from the transmitter to the receiver and $NL_t L_r$ paths involving the reconfigurable surface. Each of the latter paths has an attenuation $\alpha_{r,n,j}\alpha_{t,n,i}$ that is the product of the attenuation from the transmitter to the metaatom and from the metaatom to the receiver. Each such path is also associated with a phase-shift $e^{-j2\pi f_c(\tau_{r,n,j} + \tau_{t,n,i} + \tau_{\psi_n} - \eta)}$ containing the accumulated delays. The sinc-function determines how the signal energy carried by the path is divided between the $T + 1$ channel taps.

When the reconfigurable surface is in the far-field of the transmitter, receiver, and other objects in the propagation environment, we can represent the channel using array response vectors. We assume that the surface is a UPA with N_V rows and N_H metaatoms per row. The i th incident path to the surface can be associated with an angle pair $(\varphi_{i,i}, \theta_{i,i})$, measured from the broadside direction of the surface, and the j th outgoing path can be associated with an angle pair $(\varphi_{o,j}, \theta_{o,j})$. If we gather the N phase-shifts related to such a path, they match with the array response vector expression in (4.128):

$$\mathbf{a}_{N_H, N_V}(\varphi_{i,i}, \theta_{i,i}) = \begin{bmatrix} 1 \\ e^{-j2\pi f_c(\tau_{t,2,i} - \tau_{t,1,i})} \\ \vdots \\ e^{-j2\pi f_c(\tau_{t,N,i} - \tau_{t,1,i})} \end{bmatrix}, \quad (9.62)$$

$$\mathbf{a}_{N_H, N_V}(\varphi_{o,j}, \theta_{o,j}) = \begin{bmatrix} 1 \\ e^{-j2\pi f_c(\tau_{r,2,j} - \tau_{r,1,j})} \\ \vdots \\ e^{-j2\pi f_c(\tau_{r,N,j} - \tau_{r,1,j})} \end{bmatrix}. \quad (9.63)$$

Furthermore, the attenuation is the same for all the metaatoms: $\alpha_{t,n,i} = \alpha_{t,1,i}$ and $\alpha_{r,n,j} = \alpha_{r,1,j}$ for all n , where we take the first metaatom as the reference. For a given path, the delay variations across the surface are negligible in the sense that $B\tau_{r,n,j} \approx B\tau_{r,1,j}$ and $B\tau_{t,n,i} \approx B\tau_{t,1,i}$, for all n . We can utilize these properties to rewrite the channel coefficients in (9.61) as

$$h_{\psi}[\ell] = c_s[\ell] + \sum_{j=1}^{L_r} \sum_{i=1}^{L_t} c_{i,j}[\ell] \mathbf{a}_{N_H, N_V}^T(\varphi_{o,j}, \theta_{o,j}) \mathbf{D}_{\psi} \mathbf{a}_{N_H, N_V}(\varphi_{i,i}, \theta_{i,i}), \quad (9.64)$$

where $\mathbf{D}_{\psi} = \text{diag}(e^{j\psi_1}, \dots, e^{j\psi_N})$ contains the controllable phase-shifts

$$\psi_n = -2\pi f_c \tau_{\psi_n} \quad (9.65)$$

created by each of the metaatoms, and the channel coefficients that depend on the tap index are gathered in the sequences

$$c_s[\ell] = \sum_{i=1}^{L_s} \alpha_{s,i} e^{-j2\pi f_c(\tau_{s,i} - \eta)} \text{sinc}(\ell + B(\eta - \tau_{s,i})), \quad (9.66)$$

$$c_{i,j}[\ell] = \alpha_{r,1,j} \alpha_{t,1,i} e^{-j2\pi f_c(\tau_{r,1,j} + \tau_{t,1,i} - \eta)} \text{sinc}(\ell + B(\eta - \tau_{r,1,j} - \tau_{t,1,i})). \quad (9.67)$$

The channel coefficient at the ℓ th tap has the same structure as the narrowband channel expression in (9.23), but the fact that there are multiple taps calls for a different kind of signal transmission. If we apply OFDM with S data symbols per block and a cyclic prefix of length T , it follows from (7.25) that we obtain S memoryless subcarriers of the kind

$$\bar{y}[\nu] = \bar{h}_\psi[\nu]\bar{\chi}[\nu] + \bar{n}[\nu], \quad \text{for } \nu = 0, \dots, S-1, \quad (9.68)$$

with the reconfigurable frequency-domain channel coefficients

$$\bar{h}_\psi[\nu] = \bar{c}_s[\nu] + \sum_{j=1}^{L_r} \sum_{i=1}^{L_t} \bar{c}_{i,j}[\nu] \mathbf{a}_{N_H, N_V}^T(\varphi_{o,j}, \theta_{o,j}) \mathbf{D}_\psi \mathbf{a}_{N_H, N_V}(\varphi_{i,i}, \theta_{i,i}) \quad (9.69)$$

that depend on the DFTs of the time-domain channel coefficients:

$$\bar{c}_s[\nu] = \sum_{\ell=0}^T c_s[\ell] e^{-j2\pi\ell\nu/S}, \quad \nu = 0, \dots, S-1, \quad (9.70)$$

$$\bar{c}_{i,j}[\nu] = \sum_{\ell=0}^T c_{i,j}[\ell] e^{-j2\pi\ell\nu/S}, \quad \nu = 0, \dots, S-1. \quad (9.71)$$

The expression in (9.69) might seem complicated but can be expressed as

$$\bar{h}_\psi[\nu] = \underbrace{\begin{bmatrix} 1 \\ e^{j\psi_1} \\ \vdots \\ e^{j\psi_N} \end{bmatrix}}_{=\boldsymbol{\psi}^T} \underbrace{\begin{bmatrix} \bar{c}_s[\nu] \\ \sum_{j=1}^{L_r} \sum_{i=1}^{L_t} \bar{c}_{i,j}[\nu] \mathbf{a}_{N_H, N_V}(\varphi_{o,j}, \theta_{o,j}) \odot \mathbf{a}_{N_H, N_V}(\varphi_{i,i}, \theta_{i,i}) \end{bmatrix}}_{=\check{\mathbf{h}}[\nu]}, \quad (9.72)$$

where $\odot$ denotes the entry-wise product between two vectors. The expression $\bar{h}_\psi[\nu] = \boldsymbol{\psi}^T \check{\mathbf{h}}[\nu]$ is the simplest we can obtain for the subcarrier channel coefficient when using a reconfigurable surface. It is the inner product between $\boldsymbol{\psi}^*$ and $\check{\mathbf{h}}[\nu]$, where the former vector depends on the surface configuration, while the latter fully characterizes the channel on subcarrier ν .

9.3.3 Wideband Capacity Maximization

The capacity of a SISO-OFDM system was presented in Theorem 7.1 with arbitrary but static channel coefficients. For a given surface configuration $\boldsymbol{\psi}$, as defined in (9.72), the capacity becomes

$$C = \frac{B}{T+S} \sum_{\nu=0}^{S-1} \log_2 \left(1 + \frac{q_\nu^{\text{opt}} |\boldsymbol{\psi}^T \check{\mathbf{h}}[\nu]|^2}{N_0} \right) \quad \text{bit/s}, \quad (9.73)$$

where T is the length of the cyclic prefix,

$$q_{\nu}^{\text{opt}} = \max \left(\mu - \frac{N_0}{|\boldsymbol{\psi}^T \check{\mathbf{h}}[\nu]|^2}, 0 \right), \quad \nu = 0, \dots, S-1, \quad (9.74)$$

and the variable μ is selected to make $\sum_{\nu=0}^{S-1} q_{\nu}^{\text{opt}} = qS$.

The capacity value in (9.73) depends on the configuration $\boldsymbol{\psi}$. Corollary 9.1 showed which configuration maximizes the SNR in the narrowband case, which also leads to the maximum capacity in that case. The solution was to select $\boldsymbol{\psi}$ so that the $N+1$ terms in the inner product $\boldsymbol{\psi}^T \check{\mathbf{h}}$, where $\check{\mathbf{h}}$ was defined in (9.38), get the same phase. The optimization task is more challenging in the wideband OFDM scenario because there are S different SNRs among the S subcarriers. The SNR on subcarrier ν is proportional to $|\boldsymbol{\psi}^T \check{\mathbf{h}}[\nu]|^2$, where the channel vector $\check{\mathbf{h}}[\nu]$ is subcarrier-dependent while the surface configuration vector $\boldsymbol{\psi}$ is not. This resembles the analog beamforming situation in Section 7.3.1, where the same beamforming vector had to be used on all subcarriers. The wideband capacity maximization problem is computationally challenging but is partially addressed in [164], [165]. In this section, we will cover a suboptimal but effective way to configure the surface in these situations.

The optimal surface configuration in the narrowband case maximizes the channel gain. We can aim to do the same in the wideband case by maximizing the total channel gain over all subcarriers:

$$\sum_{\nu=0}^{S-1} |\bar{h}_{\boldsymbol{\psi}}[\nu]|^2 = \sum_{\nu=0}^{S-1} |\boldsymbol{\psi}^T \check{\mathbf{h}}[\nu]|^2 = \boldsymbol{\psi}^H \underbrace{\left(\sum_{\nu=0}^{S-1} \check{\mathbf{h}}^*[\nu] \check{\mathbf{h}}^T[\nu] \right)}_{=\mathbf{A}} \boldsymbol{\psi}. \quad (9.75)$$

If we could pick $\boldsymbol{\psi}$ as any unit-norm vector, this quadratic form would be maximized by selecting $\boldsymbol{\psi}$ as the dominant eigenvector of $\mathbf{A} = \sum_{\nu=0}^{S-1} \check{\mathbf{h}}^*[\nu] \check{\mathbf{h}}^T[\nu]$ (associated with the largest positive eigenvalue). However, we have a stricter constraint on the configuration vector because the first entry must be 1, and the remaining ones must have unit magnitude. There is no simple way to maximize (9.75) under this constraint, but an efficient iterative algorithm was proposed in [166]. The starting point is the *power iteration method* [167], which finds the dominant eigenvector of $\mathbf{A}$ by the iterative computation

$$\mathbf{w}_{i+1} = \frac{\mathbf{A}\mathbf{w}_i}{\|\mathbf{A}\mathbf{w}_i\|}, \quad i = 0, 1, \dots, \quad (9.76)$$

which is initialized from arbitrary non-zero vector $\mathbf{w}_0 \in \mathbb{C}^{N+1}$. In each iteration, the multiplication $\mathbf{A}\mathbf{w}_i$ amplifies the component of $\mathbf{w}_i$ that is aligned with the dominant eigenvector relative to all other components. The

convergence speed of the power iteration method depends on how much larger the largest eigenvalue is compared to the second largest eigenvalue.

A modified power iteration is described in Algorithm 9.1, which is initialized from $\boldsymbol{\psi}_0 = [1, \dots, 1]^T$, where the surface is not changing the phases. In each iteration, the computation in (9.76) is made using $\mathbf{w}_i = \boldsymbol{\psi}_i$. The result is used to determine the next surface configuration $\boldsymbol{\psi}_{i+1}$ by only keeping the phases of the entries of $\mathbf{w}_{i+1}$ while replacing their magnitudes by 1. Since the same value is obtained in (9.75) for $\boldsymbol{\psi}$ and $e^{-j\phi}\boldsymbol{\psi}$, for any common phase-shift ϕ , we can shift the phase of all the entries so that the first entry in $\boldsymbol{\psi}_{i+1}$ becomes 1. This shift is necessary since the phase of the static channel cannot be modified. Note that $[\mathbf{w}]_n$ denotes the n th entry of $\mathbf{w}$ in the algorithm.

Example 9.7. Suppose there is only one path to and from the reconfigurable surface (i.e., $L_t = L_r = 1$). Which phase-shift configuration maximizes (9.75)?

Under these conditions, the channel vector in (9.72) simplifies to

$$\check{\mathbf{h}}[\nu] = \begin{bmatrix} \bar{c}_s[\nu] \\ \bar{c}[\nu]\mathbf{a} \end{bmatrix}, \quad (9.77)$$

where $\mathbf{a} = \mathbf{a}_{N_H, N_V}(\varphi_o, \theta_o) \odot \mathbf{a}_{N_H, N_V}(\varphi_i, \theta_i)$ and we dropped the path indices. Hence, the matrix $\mathbf{A}$ in (9.75) can be expressed as

$$\begin{aligned} \mathbf{A} &= \sum_{\nu=0}^{S-1} \check{\mathbf{h}}^*[\nu] \check{\mathbf{h}}^T[\nu] \\ &= \sum_{\nu=0}^{S-1} \begin{bmatrix} |\bar{c}_s[\nu]|^2 & \bar{c}_s^*[\nu] \bar{c}[\nu] \mathbf{a}^T \\ \bar{c}_s[\nu] \bar{c}^*[\nu] \mathbf{a}^* & |\bar{c}[\nu]|^2 \mathbf{a}^* \mathbf{a}^T \end{bmatrix} = \begin{bmatrix} b_{ss} & b_s \mathbf{a}^T \\ b_s^* \mathbf{a}^* & b \mathbf{a}^* \mathbf{a}^T \end{bmatrix}, \end{aligned} \quad (9.78)$$

where $b_{ss} = \sum_{\nu=0}^{S-1} |\bar{c}_s[\nu]|^2 \geq 0$, $b = \sum_{\nu=0}^{S-1} |\bar{c}[\nu]|^2 \geq 0$, and $b_s = \sum_{\nu=0}^{S-1} \bar{c}_s^*[\nu] \bar{c}[\nu]$. If the phase-shift configuration is expressed as $\boldsymbol{\psi} = [1, \mathbf{w}^T]^T$, then

$$\begin{aligned} \boldsymbol{\psi}^H \mathbf{A} \boldsymbol{\psi} &= b_{ss} + b_s \mathbf{a}^T \mathbf{w} + b_s^* \mathbf{w}^H \mathbf{a}^* + b \mathbf{w}^H \mathbf{a}^* \mathbf{a}^T \mathbf{w} \\ &= b_{ss} + 2\Re\{b_s \mathbf{a}^T \mathbf{w}\} + b |\mathbf{a}^T \mathbf{w}|^2. \end{aligned} \quad (9.79)$$

Among all vectors that satisfy $\|\mathbf{w}\|^2 = N$, the third term is maximized when $\mathbf{w} = e^{-j\phi} \mathbf{a}^*$ for any ϕ , while the second term is maximized similarly but only for $\phi = \arg(b_s)$. The resulting solution $\mathbf{w} = e^{-j \arg(b_s)} \mathbf{a}^*$ is achievable with a reconfigurable surface since $\mathbf{a}$ is a vector with phase-shifts obtained from two array response vectors. Hence, the configuration that maximizes the total channel gain is parallel to the complex conjugate of the element-wise product $\mathbf{a}$ of the array response vectors to/from the surface and has an additional common phase-shift $\arg(\sum_{\nu=0}^{S-1} \bar{c}_s^*[\nu] \bar{c}[\nu])$ that aligns the static and controllable paths to the extent possible in an OFDM system.

Algorithm 9.1 Constrained power iteration to maximize (9.75).

- 1: **Initialization:** Select $\psi_0 = [1, \dots, 1]^T$ and number of iterations L
 - 2: **for** $i = 0, \dots, L - 1$ **do**
 - 3: $\mathbf{w}_{i+1} \leftarrow \frac{\mathbf{A}\psi_i}{\|\mathbf{A}\psi_i\|}$
 - 4: $\phi \leftarrow \arg([\mathbf{w}_{i+1}]_1)$
 - 5: $\psi_{i+1} = [1, e^{j(\arg([\mathbf{w}_{i+1}]_2) - \phi)}, \dots, e^{j(\arg([\mathbf{w}_{i+1}]_{N+1}) - \phi)}]^T$
 - 6: **end for**
 - 7: **Output:** ψ_L
-

To evaluate the effectiveness of the power iteration method in general propagation scenarios, we can compare the capacity that it achieves with an upper bound. Suppose we could select a different value of ψ on each subcarrier. We could then simultaneously maximize the channel gains of all subcarriers by following the approach in (9.30)–(9.31) in the narrowband case. The resulting upper bound on the capacity can be expressed as

$$C \leq \frac{B}{T+S} \sum_{\nu=0}^{S-1} \log_2 \left(1 + \frac{q_\nu \|\check{\mathbf{h}}[\nu]\|_1^2}{N_0} \right), \quad (9.80)$$

where $\|\cdot\|_1$ denotes the 1-norm and $q_\nu = \max(\mu - N_0/\|\check{\mathbf{h}}[\nu]\|_1^2, 0)$, for $\nu = 0, \dots, S-1$, with μ selected to make $\sum_{\nu=0}^{S-1} q_\nu = qS$. The upper bound is only achieved with equality in the unlikely event that the same surface configuration happens to maximize the channel gains on all subcarriers simultaneously.

Figure 9.19 shows how the capacity varies with the bandwidth in a scenario of the kind illustrated in Figure 9.10. Specifically, a reconfigurable surface is deployed along the yz -plane with its reflective side facing the positive x -axis and its center at $(0, 0, 0)$ m. The base station and the user are located at $(40, -200, 0)$ m and $(20, 0, 0)$ m, respectively. We assume LOS channels with multiple reflected paths to and from the surface, while the static channel is of NLOS nature. The surface has the size 0.5×0.5 m, which for a carrier frequency of 3 GHz corresponds to $N = 400$ metaatoms that each has the dimension $\lambda/4 \times \lambda/4$. The channels are modeled similarly to the 3GPP channel model in [168], and the capacity is averaged over random realizations of the multipath components' characteristics. The capacity in (9.73) is shown in Figure 9.19 as a function of the bandwidth B , assuming that the transmit power grows proportionally to the bandwidth so that the signal power spectral density is 1 W per MHz. The subcarrier spacing is 150 kHz, so the number of subcarriers increases with B as well as the number of channel taps.

The dashed-dotted curve in Figure 9.19 uses the power iteration method, and it provides 96–99% of the upper bound from (9.80). This method will find a configuration that takes the signal power from the strongest incident direction

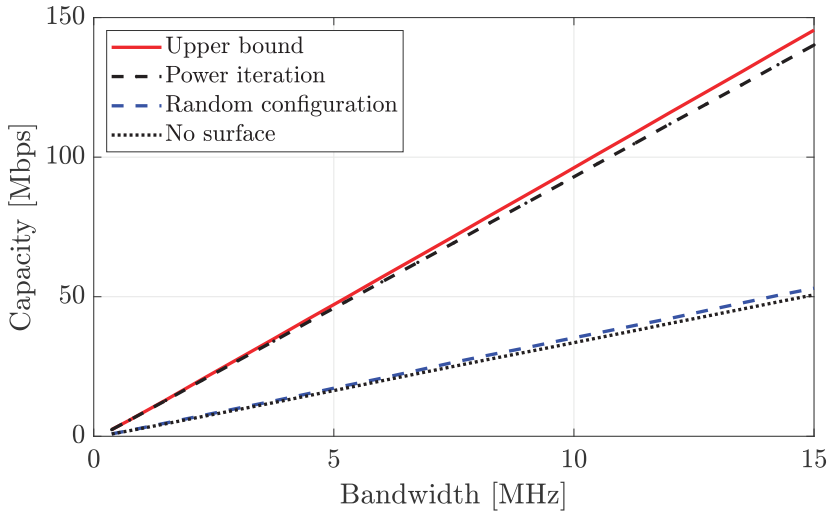


Figure 9.19: The capacity achieved over a wideband OFDM channel grows proportionally to the bandwidth and can be improved using a reconfigurable surface. The capacity achieved when configuring the surface using the power iteration in Algorithm 9.1 is compared with the upper bound in (9.80), the use of a random configuration, and the removal of the surface.

and reflects it in the direction that maximizes the received signal power. Since there are LOS channels to and from the surface, this is approximately equal to taking the signal from the LOS path and beamform it along the LOS path to the receiver. The performance gap grows with B due to the increased frequency-selectivity, but since the LOS paths to/from the surface are stronger than the scattered paths, it is possible to find a single surface configuration that works well over the entire band. This is reminiscent of how the analog beamforming architecture can provide rates close to the capacity in LOS-dominant scenarios. A part of the gap between the power iteration method and the upper bound can be closed using a more advanced configuration algorithm (examples are given in [164], [165]), but at the expense of increased computational complexity.

The importance of properly configuring the surface is also illustrated in Figure 9.19. The dashed curve considers the average capacity over random phase-shift configurations with independent uniformly distributed phases from $[-\pi, \pi)$, while the dotted curve considers the absence of a surface (i.e., it is replaced by an absorbing material). There is barely any difference between these curves, but there is a huge performance gap compared to the power iteration method. Hence, deploying a reflecting surface in this setup is only meaningful if it is configured to beamform the signal toward the receiver. The capacity is increased by 2–3 times when doing that, which results in large bit rate differences when many MHz of bandwidth are used.

9.4 MIMO Applications of Reconfigurable Surfaces

Reconfigurable surfaces can also be used to enhance MIMO channels, but there are limitations to what can be achieved since all signals reflected by a particular metaatom are phase-shifted identically. To shed light on the fundamentals using simple notation, we return to the narrowband case in this section and consider three different scenarios: point-to-point MIMO and multi-user MIMO communications, as well as multi-antenna target detection.

9.4.1 Enhanced Point-to-Point MIMO Communication

In the point-to-point SISO case considered previously in this chapter, the end-to-end channel coefficient was expressed in (9.23) as $h = h_s + \mathbf{h}_r^T \mathbf{D}_\psi \mathbf{h}_t$. In a point-to-point MIMO scenario with K transmit antennas and M receive antennas, the channel matrix $\mathbf{H} \in \mathbb{C}^{M \times K}$ can be expressed similarly as

$$\mathbf{H} = \mathbf{H}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t, \quad (9.81)$$

where $\mathbf{H}_s \in \mathbb{C}^{M \times K}$ is the static channel, $\mathbf{H}_t \in \mathbb{C}^{N \times K}$ is the channel from the transmitter to the surface, and $\mathbf{H}_r \in \mathbb{C}^{M \times N}$ is the channel from the surface to the receiver. These channel matrices can be modeled just as any other MIMO channels because the propagation to/from the surface is the same as if the array of metaatoms were an array of antennas.

The reflection matrix does not change in the MIMO case but is defined as in (9.17): $\mathbf{D}_\psi = \text{diag}(e^{j\psi_1}, \dots, e^{j\psi_N})$. This matrix can adjust how the matrices $\mathbf{H}_r$ and $\mathbf{H}_t$ are multiplied in (9.81), but the flexibility is limited since it contains N controllable phase parameters while there are respectively MN and KN coefficients in the channel matrices. These numbers were equal in the SISO case with $M = K = 1$, while there are many more channel coefficients than controllable parameters in the MIMO case.

When matrices are multiplied, the rank of the resulting matrix is always smaller or equal to the minimum rank of the individual matrices. The reflection matrix always has full rank. However, the rank of $\mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t$ cannot surpass the minimum rank of the channel matrices $\mathbf{H}_r$ and $\mathbf{H}_t$. This implies that the reconfigurable surface cannot improve the channel rank in any dramatic fashion. However, it can be configured to improve specific singular values, match the strongest channel dimensions from the two channel matrices, and ensure that the static and configurable terms in (9.81) fit well together. Since many possibilities exist, we should identify the surface configuration that maximizes the MIMO channel capacity. In this section, we will first provide some geometrical insights into what can be achieved and then derive a general algorithm that iteratively refines the configuration to increase capacity.

Example 9.8. Suppose the surface is deployed to have far-field LOS channels to both the transmitter and receiver. How should the surface be configured to maximize the total end-to-end channel gain $\|\mathbf{H}\|_F^2$?

The matrices $\mathbf{H}_r$ and $\mathbf{H}_t$ have rank one under these conditions, as explained in Section 4.4.1. They could be expressed as outer products of array response vectors, but for notational convenience, we will express them as

$$\mathbf{H}_r = \mathbf{a}_r \mathbf{b}_r^T, \quad \mathbf{H}_t = \mathbf{a}_t \mathbf{b}_t^T, \quad (9.82)$$

where $\mathbf{a}_r \in \mathbb{C}^M$, $\mathbf{b}_r = [b_{r,1}, \dots, b_{r,N}]^T \in \mathbb{C}^N$, $\mathbf{a}_t = [a_{t,1}, \dots, a_{t,N}]^T \in \mathbb{C}^N$, and $\mathbf{b}_t \in \mathbb{C}^K$ are vectors. We can then simplify (9.81) as

$$\mathbf{H} = \mathbf{H}_s + \underbrace{\mathbf{a}_r \mathbf{b}_r^T \mathbf{D}_\psi \mathbf{a}_t \mathbf{b}_t^T}_{=\alpha}, \quad (9.83)$$

where $\alpha = \mathbf{b}_r^T \mathbf{D}_\psi \mathbf{a}_t \in \mathbb{C}$ is a scalar. The second term adds the rank-one matrix $\mathbf{a}_r \mathbf{b}_t^T$ to the static channel with a scaling that is determined by the reflection matrix. The total end-to-end channel gain can be rewritten using (5.88) as

$$\begin{aligned} \|\mathbf{H}\|_F^2 &= \text{tr}(\mathbf{H}^H \mathbf{H}) \\ &= \text{tr}(\mathbf{H}_s^H \mathbf{H}_s) + |\alpha|^2 \text{tr}(\mathbf{b}_t^* \mathbf{a}_r^H \mathbf{a}_r \mathbf{b}_t) + \text{tr}(\alpha \mathbf{H}_s^H \mathbf{a}_r \mathbf{b}_t^T) + \text{tr}(\alpha^* \mathbf{b}_t^* \mathbf{a}_r^H \mathbf{H}_s) \\ &= \|\mathbf{H}_s\|_F^2 + |\alpha|^2 \|\mathbf{a}_r\|^2 \|\mathbf{b}_t\|^2 + 2\Re(\alpha \mathbf{b}_t^T \mathbf{H}_s^H \mathbf{a}_r), \end{aligned} \quad (9.84)$$

where the last equality follows from (2.52) that states how one can shift the order of matrices in the trace function. The final expression is maximized when $|\alpha|$ takes its largest possible value while its phase makes the last term positive. This happens when $\psi_n = -\arg(\mathbf{b}_t^T \mathbf{H}_s^H \mathbf{a}_r) - \arg(b_{r,n} a_{t,n}) + 2\pi k_n$, where k_n is the integer that gives $\psi_n \in [-\pi, \pi)$, for $n = 1, \dots, N$. When the static channel is weak, maximizing $\|\mathbf{H}\|_F^2$ is approximately equivalent to maximizing the channel capacity because there will only be one strong singular value, which is amplified using the surface.

As noted earlier in this chapter, deploying the reconfigurable surface to have LOS channels to the transmitter and receiver is desirable. However, in contrast to the example, the channel matrices will not have rank one in practice due to multipath propagation. This makes it harder to compute the optimal phase-shift configuration directly. The design principle remains the same, as can be showcased using the beamspace representation. Suppose the transmitter and receiver are equipped with half-wavelength-spaced ULAs. As explained in Section 5.6.2, we can then transform the channel matrix to the beamspace by multiplying by DFT matrices from the left and the right:

$$\check{\mathbf{H}} = \mathbf{F}_M^H \mathbf{H} \mathbf{F}_K^* = \mathbf{F}_M^H \mathbf{H}_s \mathbf{F}_K^* + \mathbf{F}_M^H \mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t \mathbf{F}_K^*. \quad (9.85)$$

The first term represents all the static multipath clusters, while the second

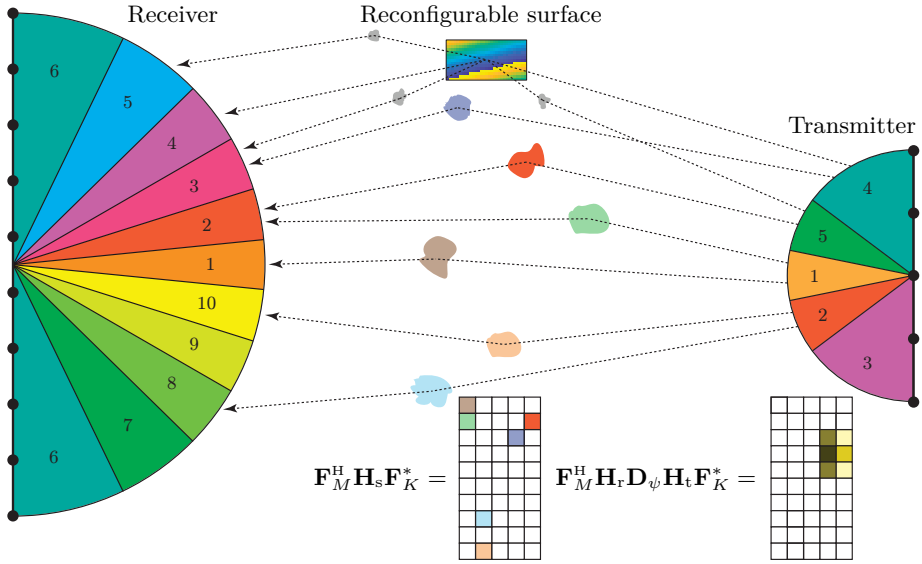


Figure 9.20: A transmitter with a half-wavelength-spaced ULA with $K = 5$ antennas communicates with a receiver having a half-wavelength-spaced ULA with $M = 10$ antennas. The communication is aided by a reconfigurable surface that adds extra paths to the MIMO channel. In the beamspace representation, these extra paths are concentrated around a few angular directions since the surface mostly interacts with multipath components in its vicinity. This is a continuation of Figure 5.33 where no reconfigurable surfaces existed. Note that the transmitter and receiver sizes are exaggerated compared to the propagation distances.

term represents the propagation paths affected by the reconfigurable surface. Figure 9.20 exemplifies how these matrices depend on the angular geometry and is a continuation of Figure 5.33, which considered the same setup without a reconfigurable surface. Each of the six static multipath clusters contributes to the entry of $\mathbf{F}_M^H \mathbf{H}_s \mathbf{F}_K^*$ with the matching color. A white entry means its value is nearly zero because no propagation path connects that pair of transmit/receive directions. Since we consider deployment with LOS to the transmitter and receiver, the reconfigurable surface mainly contributes to one entry, determined by its physical location. However, it could slightly affect a few neighboring entries using the multipath clusters around it, as illustrated by the brightness of the coloring (brighter means smaller).

The rank of $\mathbf{H}$ is the same as the rank of $\tilde{\mathbf{H}}$. The static channel matrix describes the contributions from all multipath clusters that create paths between the transmitter and the receiver. Clusters seen from very different angles barely interact and contribute to different singular values in $\tilde{\mathbf{H}}$. The reconfigurable surface can increase the channel rank by creating new non-zero entries. Since the surface is deployed at a specific location, it will primarily interact with the propagation environment in its vicinity. Hence, all the new propagation paths it creates have similar angles from the transmitter's and receiver's perspectives.

This implies that we should expect the surface to mainly contribute to one or a few singular values in $\tilde{\mathbf{H}}$. The reconfigurable surface can raise the channel capacity by selecting a good phase-shift configuration, but the increase will resemble the SISO case since only one channel dimension is improved. Multiple reconfigurable surfaces deployed at physically diverse locations are generally required to enhance multiple singular values substantially. An analogy can be made with the analog/hybrid beamforming considered in Sections 7.3.1 and 7.3.2: a single surface is like an analog beamforming architecture that can only receive/transmit one signal, while multiple surfaces are like the hybrid beamforming architecture that can jointly receive/transmit as many signals as there are surfaces. While this is generally true, there are specific scenarios where the surface contributes to many singular values [169]. This happens when the arrays and surface are very large compared to the wavelength so that spherical wavefronts can be utilized to achieve high-rank channels via the surface. The enabling factor is the same as in Section 4.4.3, where we showed how to achieve full-rank LOS channels by making the antenna spacing sufficiently large compared to the propagation distance. Similarly, one can make the reconfigurable surface so large compared to the propagation distances that it acts as multiple surfaces.

For a given MIMO channel matrix, the capacity is achieved by transmitting along the right singular vectors and allocating power using water-filling (see Theorem 3.1). If we modify the channel matrix by refining the reflection matrix $\mathbf{D}_\psi$, the singular vectors and values will change, and so will the capacity-achieving precoding. We will describe an algorithm from [170] that progressively improves the MIMO capacity by iterating between updating the precoding/water-filling and reflection matrix. We recall from (3.100) that the capacity for a given channel matrix $\mathbf{H}$ can be expressed as

$$C = \log_2 \left(\det \left(\mathbf{I}_M + \frac{1}{N_0} \mathbf{H} \mathbf{V} \mathbf{Q}^{\text{opt}} \mathbf{V}^H \mathbf{H}^H \right) \right). \quad (9.86)$$

We introduce the notation $\mathbf{H}_r = [\mathbf{h}_{r,1}, \dots, \mathbf{h}_{r,N}]$ and $\mathbf{H}_t = [\vec{\mathbf{h}}_{t,1}, \dots, \vec{\mathbf{h}}_{t,N}]^T$, where $\vec{\mathbf{h}}_{t,n}^T$ is the n th row and the arrow notation points out that rows are horizontal. We can then rewrite the channel matrix in (9.81) as

$$\mathbf{H} = \mathbf{H}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t = \mathbf{H}_s + \sum_{i=1}^N \mathbf{h}_{r,i} e^{j\psi_i} \vec{\mathbf{h}}_{t,i}^T = \mathbf{H}_n + e^{j\psi_n} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T, \quad (9.87)$$

where $\mathbf{H}_n = \mathbf{H}_s + \sum_{i=1, i \neq n}^N \mathbf{h}_{r,i} e^{j\psi_i} \vec{\mathbf{h}}_{t,i}^T$ contains all the terms except the one involving ψ_n . We want to reconfigure this phase-shift to increase the capacity when all other parameters are fixed. By substituting (9.87) into (9.86), we

can express the capacity as

$$\begin{aligned}
 & \log_2 \left(\det \left(\mathbf{I}_M + \frac{1}{N_0} \left(\mathbf{H}_n + e^{j\psi_n} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T \right) \mathbf{V} \mathbf{Q}^{\text{opt}} \mathbf{V}^H \left(\mathbf{H}_n + e^{j\psi_n} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T \right)^H \right) \right) \\
 &= \log_2 \left(\det \left(\mathbf{A}_n + e^{j\psi_n} \mathbf{h}_{r,n} \mathbf{b}_n^H + e^{-j\psi_n} \mathbf{b}_n \mathbf{h}_{r,n}^H \right) \right), \\
 &= \log_2 \left(\det (\mathbf{A}_n) \right) + \log_2 \left(\det \left(\mathbf{I}_M + e^{j\psi_n} \mathbf{A}_n^{-1} \mathbf{h}_{r,n} \mathbf{b}_n^H + e^{-j\psi_n} \mathbf{A}_n^{-1} \mathbf{b}_n \mathbf{h}_{r,n}^H \right) \right), \tag{9.88}
 \end{aligned}$$

where the terms that are independent of ψ_n are included in

$$\mathbf{A}_n = \mathbf{I}_M + \frac{1}{N_0} \mathbf{H}_n \mathbf{V} \mathbf{Q}^{\text{opt}} \mathbf{V}^H \mathbf{H}_n^H + \frac{1}{N_0} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T \mathbf{V} \mathbf{Q}^{\text{opt}} \mathbf{V}^H \vec{\mathbf{h}}_{t,n}^* \mathbf{h}_{r,n}^H, \tag{9.89}$$

$$\mathbf{b}_n = \frac{1}{N_0} \mathbf{H}_n \mathbf{V} \mathbf{Q}^{\text{opt}} \mathbf{V}^H \vec{\mathbf{h}}_{t,n}^*. \tag{9.90}$$

Only the determinant in the second term of (9.88) depends on the phase-shift. This determinant can be computed as

$$\begin{aligned}
 & \det \left(\mathbf{I}_M + [\mathbf{A}_n^{-1} \mathbf{h}_{r,n} \quad \mathbf{A}_n^{-1} \mathbf{b}_n] \begin{bmatrix} e^{j\psi_n} \mathbf{b}_n^H \\ e^{-j\psi_n} \mathbf{h}_{r,n}^H \end{bmatrix} \right) \\
 &= \det \left(\mathbf{I}_2 + \begin{bmatrix} e^{j\psi_n} \mathbf{b}_n^H \\ e^{-j\psi_n} \mathbf{h}_{r,n}^H \end{bmatrix} [\mathbf{A}_n^{-1} \mathbf{h}_{r,n} \quad \mathbf{A}_n^{-1} \mathbf{b}_n] \right) \\
 &= (1 + e^{j\psi_n} \mathbf{b}_n^H \mathbf{A}_n^{-1} \mathbf{h}_{r,n}) (1 + e^{-j\psi_n} \mathbf{h}_{r,n}^H \mathbf{A}_n^{-1} \mathbf{b}_n) - \mathbf{b}_n^H \mathbf{A}_n^{-1} \mathbf{b}_n \mathbf{h}_{r,n}^H \mathbf{A}_n^{-1} \mathbf{h}_{r,n} \\
 &= e^{j\psi_n} \mathbf{b}_n^H \mathbf{A}_n^{-1} \mathbf{h}_{r,n} + e^{-j\psi_n} \mathbf{h}_{r,n}^H \mathbf{A}_n^{-1} \mathbf{b}_n + \text{constants}, \tag{9.91}
 \end{aligned}$$

where the first equality follows from Sylvester's determinant theorem in (2.53), and we then compute the determinant for the resulting 2×2 matrix. The final expression in (9.91) is maximized when the first two terms are positive, which is achieved by

$$\psi_n = -\arg(\mathbf{b}_n^H \mathbf{A}_n^{-1} \mathbf{h}_{r,n}). \tag{9.92}$$

We can utilize this result to obtain the iterative procedure described in Algorithm 9.2. The algorithm begins by computing the capacity-achieving signal covariance matrix $\mathbf{V} \mathbf{Q}^{\text{opt}} \mathbf{V}^H$ for an initial set of phase-shifts. It then refines the N phase-shifts sequentially using (9.92). When this is done, the capacity-achieving signal covariance matrix is recomputed for the new channel matrix obtained with the new reflection matrix, and the procedure is repeated L times. Each step in the algorithm either improves the achievable rate or keeps it fixed because we can always choose not to modify the phase. The rate will eventually converge when no further changes are beneficial. However, a consequence of sequential optimization is that the algorithm might not converge to the best possible configuration but only a locally optimal solution where one cannot further increase the capacity unless multiple phases are

Algorithm 9.2 Reconfigurable surface configuration for point-to-point MIMO capacity maximization.

- 1: **Initialization:** Set $\psi_1, \dots, \psi_N$ randomly and select the number of iterations L
 - 2: **for** $i = 1, \dots, L$ **do**
 - 3: Compute the capacity-achieving covariance matrix $\mathbf{V}\mathbf{Q}^{\text{opt}}\mathbf{V}^{\text{H}}$ for the channel matrix in (9.81) with $\mathbf{D}_\psi = \text{diag}(e^{j\psi_1}, \dots, e^{j\psi_N})$
 - 4: **for** $n = 1, \dots, N$ **do**
 - 5: Compute $\mathbf{A}_n$ in (9.89) and $\mathbf{b}_n$ in (9.90) for fixed $\psi_1, \dots, \psi_N$
 - 6: $\psi_n \leftarrow -\arg(\mathbf{b}_n^{\text{H}}\mathbf{A}_n^{-1}\mathbf{h}_{\text{r},n})$
 - 7: **end for**
 - 8: **end for**
 - 9: **Output:** $\psi_1, \dots, \psi_N$
-

updated simultaneously.⁶ The channel matrices must be estimated before running Algorithm 9.2. No extra wireless signaling is required while running the algorithm, which can be executed at the receiver. Hence, the procedure shown in Figure 9.14 can still be followed: the transmitter sends pilots while the surface switches between predefined configurations, and then the receiver computes the preferred configuration and sends it to the surface.

Figure 9.21 shows how the capacity is improved with the iteration index from Algorithm 9.2 in a point-to-point MIMO scenario with $M = K \in \{1, 2, 4, 8\}$ antennas and $N = 100$ metaatoms. The SNR of the static path is 0 dB and $\mathbf{H}_s$ has i.i.d. Rayleigh fading entries. The channel matrices $\mathbf{H}_r$ and $\mathbf{H}_t$ via the surface are subject to Rician fading with the κ -factor $\kappa = 10$ and the NLOS part having an i.i.d. Rayleigh fading distribution (see Example 5.18). The cascaded path via a single metaatom has the SNR -10 dB. The results are averaged over many channel realizations.

The iteration index 0 in Figure 9.21 represents the initial state when the phase-shifts are uniformly distributed between 0 and 2π , thereby approximating diffuse scattering. The first iteration of Algorithm 9.2 leads to a substantial capacity improvement, while only minor improvements occur in the subsequent iterations. The vertical gaps between the curves grow when comparing the first and last points on the curves. This shows that a system with more antennas benefits slightly more from having a well-configured surface, but the difference is small because the surface mainly contributes to one singular value. We use the optimal configuration from Corollary 9.1 when considering the SISO case with $M = K = 1$, which is why that curve does not vary with

⁶The initial phase-shifts determine which configuration that Algorithm 9.2 converges to. A simple way to explore if better solutions exist is to consider multiple random phase-shift initializations and compare the capacity values they converge to.

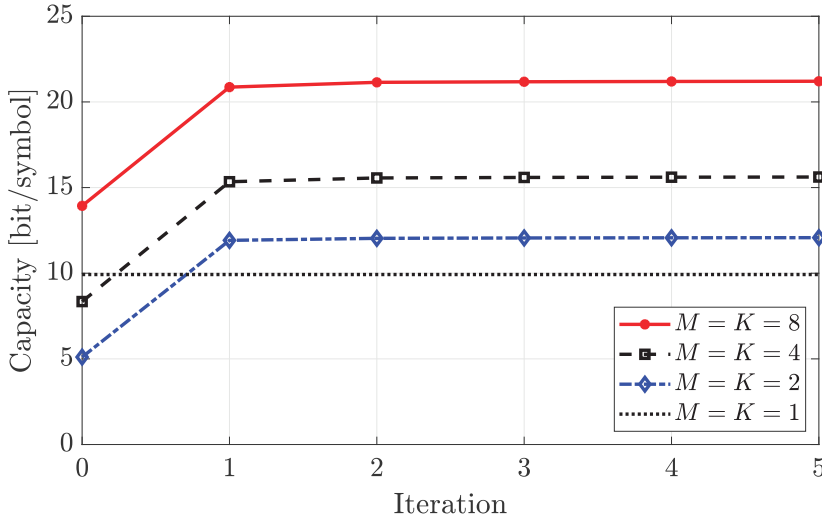


Figure 9.21: The point-to-point MIMO capacity as a function of the iteration index when running Algorithm 9.2 to iteratively select the phase-shifts of the reconfigurable surface to increase the capacity. The index 0 represents the initial random configuration. The optimal configuration is directly used in the SISO case ($M = K = 1$) since it is known in closed form.

the iteration index. Interestingly, the optimal SISO configuration leads to a higher capacity than the initial capacity in the 2×2 and 4×4 MIMO setups, reiterating the importance of correctly configuring the surface. In summary, the deployment of a reconfigurable surface can greatly improve the capacity of a point-to-point MIMO system.

9.4.2 Enhanced Multi-User MIMO Communication

A reconfigurable surface can also improve the communication performance over multi-user MIMO channels. As explained in Chapter 6, the precoding and combining differ substantially from the point-to-point case because users are not collaborating in the signal processing and measure their capacity separately. Nevertheless, the sum capacity expression in multi-user MIMO resembles the capacity expression in point-to-point MIMO, which implies that we can use similar algorithms to optimize the reconfigurable surface.

The uplink sum capacity in a multi-user MIMO system with K single-antenna users and M antennas at the base station is given in (6.49) as

$$\log_2 \left(\det \left(\mathbf{I}_M + \frac{q}{N_0} \mathbf{H} \mathbf{H}^H \right) \right) \quad \text{bit/symbol}, \quad (9.93)$$

where $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_K] \in \mathbb{C}^{M \times K}$ is the channel matrix and $q = P/B$ is the signal energy per symbol. When the uplink is aided by a reconfigurable surface

with N metaatoms, the channel $\mathbf{h}_k \in \mathbb{C}^M$ from user k can be expressed as

$$\mathbf{h}_k = \mathbf{h}_{s,k} + \mathbf{H}_r \mathbf{D}_\psi \mathbf{h}_{t,k}, \quad (9.94)$$

where $\mathbf{h}_{s,k} \in \mathbb{C}^M$ is the static channel, $\mathbf{h}_{t,k} \in \mathbb{C}^N$ is the channel from the user to the surface, and $\mathbf{H}_r \in \mathbb{C}^{M \times N}$ is the channel from the surface to the receiver. This is the SIMO counterpart to the channel model in (9.81). The term $\mathbf{H}_r \mathbf{D}_\psi \mathbf{h}_{t,k}$ in (9.94) can be viewed as the projection of the user-specific channel vector $\mathbf{h}_{t,k}$ onto the span of the matrix $\mathbf{H}_r \mathbf{D}_\psi$, which is the same for all users but controllable using the reflection matrix $\mathbf{D}_\psi = \text{diag}(e^{j\psi_1}, \dots, e^{j\psi_N})$. The combined channel matrix of all users becomes

$$\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_K] = \underbrace{[\mathbf{h}_{s,1}, \dots, \mathbf{h}_{s,K}]}_{=\mathbf{H}_s} + \mathbf{H}_r \mathbf{D}_\psi \underbrace{[\mathbf{h}_{t,1}, \dots, \mathbf{h}_{t,K}]}_{=\mathbf{H}_t}, \quad (9.95)$$

which has the same form $\mathbf{H} = \mathbf{H}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t$ as in the point-to-point MIMO case. In particular, the reflection matrix enters into the equation identically.

Example 9.9. Suppose the surface is deployed to have a far-field LOS channel to the base station. How can it modify the user channels in this case?

The matrix $\mathbf{H}_r$ has rank one under these conditions, as explained in Section 4.4.1, and can be expressed as $\mathbf{H}_r = \mathbf{a}_r \mathbf{b}_r^T$ for some vectors $\mathbf{a}_r \in \mathbb{C}^M$ and $\mathbf{b}_r \in \mathbb{C}^N$. The channel of user k in (9.94) then becomes

$$\mathbf{h}_k = \mathbf{h}_{s,k} + \mathbf{a}_r \underbrace{\mathbf{b}_r^T \mathbf{D}_\psi \mathbf{h}_{t,k}}_{=\alpha_k}. \quad (9.96)$$

This implies that the surface adds a component $\alpha_k \mathbf{a}_r$ to the static channel vector, where only the complex scaling factor α_k can be controlled and depends on the user index. In case the static channels are blocked (i.e., $\mathbf{h}_{s,k} = \mathbf{0}$ for all k), the K channel vectors are parallel. We cannot suppress interference under such circumstances; thus, FDMA achieves the same sum capacity as multi-user MIMO in this case. In conclusion, the reconfigurable surface cannot enable multi-user MIMO communications on its own, but it can improve performance by making the channels $\mathbf{h}_k$ more diverse than the original static channels by adding the components $\alpha_k \mathbf{a}_r$.

Since there are K user capacities to consider in multi-user MIMO systems, different phase-shift configurations are preferred for different users. In other words, the reconfigurable surface bends the shape of the capacity region, and there is typically no configuration that results in a region that is larger than all other achievable regions in all user dimensions. In this section, we will concentrate on maximizing the sum capacity in (9.93). Similarly to the last section, we will develop an iterative algorithm that updates one of the N phase-shifts at a time to increase the capacity. When refining the phase ψ_n of

metaatom n , it is convenient to express the channel matrix $\mathbf{H} = \mathbf{H}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t$ in (9.95) as

$$\mathbf{H} = \mathbf{H}_s + \sum_{i=1}^N \mathbf{h}_{r,i} e^{j\psi_i} \vec{\mathbf{h}}_{t,i}^T = \mathbf{H}_s + \underbrace{\sum_{i=1, i \neq n}^N \mathbf{h}_{r,i} e^{j\psi_i} \vec{\mathbf{h}}_{t,i}^T + e^{j\psi_n} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T}_{=\mathbf{H}_n}, \quad (9.97)$$

using the notation $\mathbf{H}_r = [\mathbf{h}_{r,1}, \dots, \mathbf{h}_{r,N}]$ and $\mathbf{H}_t = [\vec{\mathbf{h}}_{t,1}, \dots, \vec{\mathbf{h}}_{t,N}]^T$. Note that $\vec{\mathbf{h}}_{t,n}^T$ is the n th row of $\mathbf{H}_t$ and differs from the user channel vector $\mathbf{h}_{t,k}$ appearing as the k th column of the matrix. By substituting (9.97) into (9.93), we can express the sum capacity as

$$\begin{aligned} & \log_2 \left(\det \left(\mathbf{I}_M + \frac{q}{N_0} \left(\mathbf{H}_n + e^{j\psi_n} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T \right) \left(\mathbf{H}_n + e^{j\psi_n} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T \right)^H \right) \right) \\ &= \log_2 \left(\det \left(\mathbf{A}_n + e^{j\psi_n} \mathbf{h}_{r,n} \mathbf{b}_n^H + e^{-j\psi_n} \mathbf{b}_n \mathbf{h}_{r,n}^H \right) \right), \\ &= \log_2 (\det (\mathbf{A}_n)) + \log_2 \left(\det \left(\mathbf{I}_M + e^{j\psi_n} \mathbf{A}_n^{-1} \mathbf{h}_{r,n} \mathbf{b}_n^H + e^{-j\psi_n} \mathbf{A}_n^{-1} \mathbf{b}_n \mathbf{h}_{r,n}^H \right) \right), \end{aligned} \quad (9.98)$$

where the terms that are independent of ψ_n are included in

$$\mathbf{b}_n = \frac{q}{N_0} \mathbf{H}_n \vec{\mathbf{h}}_{t,n}^*, \quad \mathbf{A}_n = \mathbf{I}_M + \frac{q}{N_0} \mathbf{H}_n \mathbf{H}_n^H + \frac{q}{N_0} \mathbf{h}_{r,n} \vec{\mathbf{h}}_{t,n}^T \vec{\mathbf{h}}_{t,n}^* \mathbf{h}_{r,n}^H. \quad (9.99)$$

It remains to select the phase-shift to maximize the second determinant in (9.98), and this problem has the same form as in (9.91) of the point-to-point MIMO case. Hence, the optimal phase is obtained from (9.92) as

$$\psi_n = -\arg(\mathbf{b}_n^H \mathbf{A}_n^{-1} \mathbf{h}_{r,n}), \quad (9.100)$$

but using the expressions for $\mathbf{b}_n$ and $\mathbf{A}_n$ defined above. By sequentially updating the N phases using (9.100), we obtain Algorithm 9.3. This algorithm resembles Algorithm 9.2 for the point-to-point MIMO case, but a key difference is that the precoding is not updated in multi-user MIMO because the sum capacity is always achieved when the users transmit their signals using maximum power. Each step in the algorithm either improves the sum capacity or keeps it fixed because we can always choose not to modify the phase; thus, the sum capacity gradually increases and converges to a final value. We let L denote the predefined number of iterations to consider, but the algorithm can also be terminated earlier when the sum capacity has not been improved much from one iteration to the next. Although the sum capacity improves monotonically, there is no guarantee that the algorithm will converge to the best conceivable configuration because the variables are optimized sequentially rather than jointly.

Figure 9.22 shows how the sum capacity of an uplink multi-user MIMO system increases with the number of metaatoms. There are $K = 4$ users,

Algorithm 9.3 Reconfigurable surface configuration for uplink multi-user MIMO sum capacity maximization.

- 1: **Initialization:** Set $\psi_1, \dots, \psi_N$ randomly and select the number of iterations L
 - 2: **for** $i = 1, \dots, L$ **do**
 - 3: **for** $n = 1, \dots, N$ **do**
 - 4: Compute $\mathbf{A}_n$ and $\mathbf{b}_n$ in (9.99) using current $\psi_1, \dots, \psi_N$
 - 5: $\psi_n \leftarrow -\arg(\mathbf{b}_n^H \mathbf{A}_n^{-1} \mathbf{h}_{r,n})$
 - 6: **end for**
 - 7: **end for**
 - 8: **Output:** $\psi_1, \dots, \psi_N$
-

$M = 10$ receive antennas, and the static channel is modeled as in Figure 6.16. The channel matrix $\mathbf{H}_r$ between the base station and reconfigurable surface is modeled as Rician fading with the κ -factor $\kappa = 10$ and the channels between the users and surface are subject to i.i.d. Rayleigh fading. SNR of the static path is 0 dB, while the cascaded path via a single metaatom has the SNR -20 dB. The results are averaged over many channel realizations. We notice that the sum capacity grows rapidly with the number of metaatoms when Algorithm 9.3 is used; hence, deploying the reconfigurable surface in this particular setup makes a great difference. $L = 1$ iteration of the algorithm is sufficient to outperform the initial configuration with random phase-shifts. Further capacity improvements are achieved by running $L = 5$ iterations of the algorithm, especially when there are many metaatoms to configure.

We have focused on the uplink thus far, but the results are also useful for the downlink because the uplink-downlink duality implies that we can achieve the same user rates in both directions. Hence, if the surface is configured to provide a high uplink sum capacity, we can achieve the same downlink sum rate using the same power without changing the surface configuration. However, this will generally not be the downlink sum capacity because we might have a different total downlink transmit power and can allocate it arbitrarily between the users. The downlink sum capacity was stated in (6.122) as the problem of maximizing the sum rate in the virtual uplink with respect to the virtual uplink powers, and it can be solved efficiently using convex optimization tools. It is straightforward to devise an iterative algorithm that switches between solving (6.122) for given user channels and enhancing the channels using Algorithm 9.3 for given virtual uplink powers.

The uplink sum capacity requires SIC, while the downlink sum capacity is achieved using DPC. It is easier to implement uplink and downlink multi-user MIMO systems with linear signal processing, but unfortunately, it comes at the price of more complex parameter optimization problems. For example, the

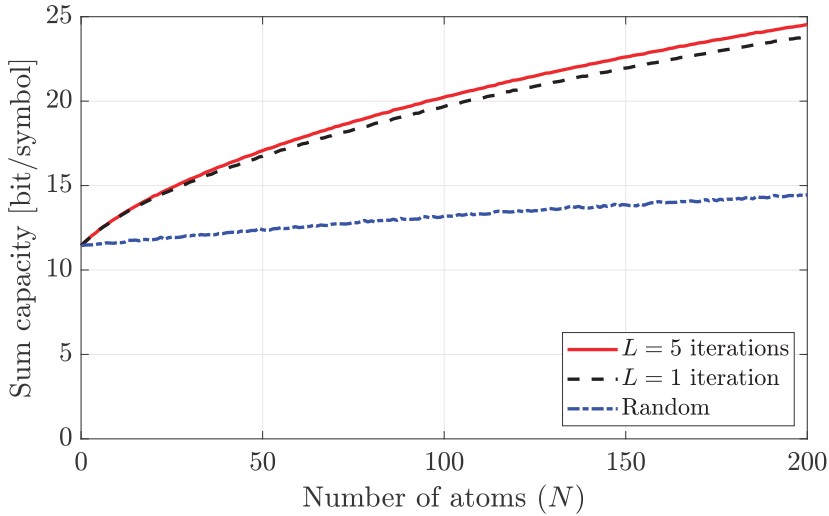


Figure 9.22: The sum capacity of an uplink multi-user MIMO system that is aided by a reconfigurable surface with a varying number of metaatoms. The phase-shift configuration is either selected randomly or by running Algorithm 9.3 with $L = 1$ or $L = 5$ iterations.

uplink sum capacity is achieved when all users transmit with their maximum power, while the maximum uplink sum rate with linear combining requires power control optimization (as exemplified in Section 6.3.6). Similarly, the parameters required to achieve the downlink sum capacity are obtained by solving the convex optimization problem in (6.122), while the linear precoding that maximizes the downlink sum rate can only be computed using high-complexity global optimization algorithms [84], [85]. The structure of the rate expressions causes increased complexity and makes phase-shift optimization more complicated when a reconfigurable surface supports a multi-user MIMO system that employs linear processing. We refer to [155], [171], [172] for further details and solutions to these problems. The bottom line is that reconfigurable surfaces can improve the user rates in multi-user MIMO systems, and many algorithms for phase-shift optimization can be developed for various utility functions and kinds of signal processing for data transmission and reception.

9.4.3 Enhanced Target Detection

A reconfigurable surface can also be used to improve the wireless channel properties for sensing applications [173], [174], particularly to increase the SNR and reliability. To exemplify this, we will consider a mono-static target detection scenario, where a multi-antenna radar system must determine whether a target exists at a specific location or not. A reconfigurable surface is deployed in the same area, and there are free-space LOS channels between the different locations, as illustrated in Figure 9.23. The radar transceiver

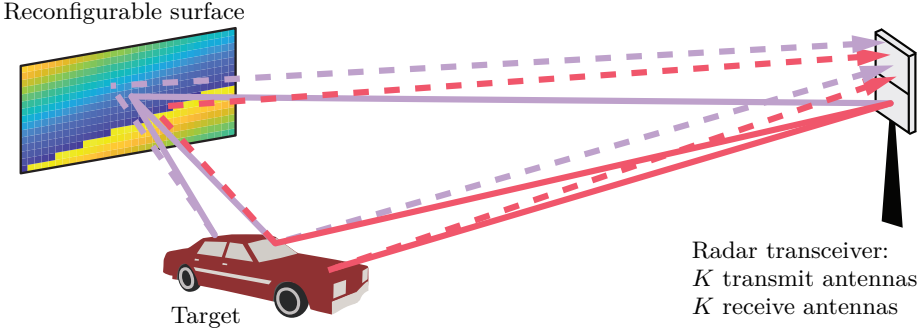


Figure 9.23: A radar transceiver with K transmit antennas and K receive antennas wants to detect the presence of a target with assistance from a reconfigurable surface with N metaatoms. There are two paths from the transmitter to the target and two paths from the target to the receiver, resulting in four propagation paths. Solid lines represent paths leading to the target and dashed lines are paths leading back to the receiver.

has K transmit antennas and K receive antennas, which are symmetrically arranged to achieve identical array response vectors. The surface consists of N metaatoms.

A predefined radar signal $\mathbf{p}x$ is transmitted using the precoding vector $\mathbf{p} \in \mathbb{C}^K$, and it reaches the target location in two ways: through the direct LOS path or via the reflection by the surface. If the target exists, it will reflect the signals, and these can reach the receiver either through the direct LOS path or via reflection by the surface. This gives rise to a total of four propagation paths from the transmitter to the receiver. We let $\mathbf{h}_s \in \mathbb{C}^K$ denote the static LOS channel between the transmitter and target location. Furthermore, the cascaded channel from the radar to the target via the reconfigurable surface is represented by the vector

$$\mathbf{h}_c = \mathbf{a}_t \mathbf{b}_t^T \mathbf{D}_\psi \mathbf{h}_r, \quad (9.101)$$

where $\mathbf{a}_t \mathbf{b}_t^T \in \mathbb{C}^{K \times N}$ is the rank-one LOS channel matrix between the radar transceiver and surface, $\mathbf{D}_\psi \in \mathbb{C}^{N \times N}$ is the reflection matrix, and $\mathbf{h}_r \in \mathbb{C}^N$ is the channel between the surface and target. For notational simplicity, we will not include any channel gains in $\mathbf{h}_s$, $\mathbf{a}_t$, $\mathbf{b}_t$, and $\mathbf{h}_r$ but model them separately. Hence, these are four array response vectors that describe the LOS propagation between the different locations, which implies that the squared norm of each vector equals the number of entries it has.

If the target exists, the effective end-to-end channel to the receiver is

$$\mathbf{h} = \underbrace{(c_1 \mathbf{h}_s \mathbf{h}_s^T)}_{\text{LOS path}} + \underbrace{c_2 \mathbf{h}_c \mathbf{h}_c^T}_{\text{Via surface}} + \underbrace{(c_3 \mathbf{h}_s \mathbf{h}_c^T + c_3 \mathbf{h}_c \mathbf{h}_s^T)}_{\text{Mix of LOS and surface paths}} \mathbf{p}, \quad (9.102)$$

where we included the precoding vector and $c_1 \sim \mathcal{N}_{\mathbb{C}}(0, \beta_1)$, $c_2 \sim \mathcal{N}_{\mathbb{C}}(0, \beta_2)$, and $c_3 \sim \mathcal{N}_{\mathbb{C}}(0, \beta_3)$ are three independent RCS realizations for the target,

which include the channel gains as well. Multiple realizations are required because we consider signals reaching and leaving the target in different directions. However, the coefficient c_3 appears twice due to channel reciprocity, which implies that the RCS is the same when the signal propagates from the transmitter to the target and back via the surface, and when the signal travels in the opposite direction. There are four terms in (9.102) representing the four propagation paths from the transmitter to the receiver. The first term is the direct reflection by the target that would also happen in the absence of the surface. The second term is the path that reaches the target via the surface and then goes back to the receiver in the same way. The third term is the path that reaches the target via the surface and then is reflected through the LOS path, while the fourth term takes the opposite direction. The variances $\beta_1, \beta_2, \beta_3$ are generally different because they include the multiplication of the channel gains between the different locations that the radar signal passes on its way from the transmitter to the receiver. We can expect that $\beta_1 > \beta_3 > \beta_2$ since the LOS path is typically stronger than the path via a single metaatom; however, with an appropriate surface configuration, the combined effect of the N metaatoms can make a large difference for target detection.

We assume that the signal $\sqrt{P}$ is transmitted, denote the received signal by $\mathbf{y} \in \mathbb{C}^K$, and formulate the binary hypothesis test

$$\mathcal{H}_0 : \mathbf{y} = \mathbf{n}, \quad (9.103)$$

$$\mathcal{H}_1 : \mathbf{y} = \sqrt{P}\mathbf{h} + \mathbf{n}, \quad (9.104)$$

where $\mathbf{n} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2 \mathbf{I}_K)$ is the additive noise vector.

If the hypothesis $\mathcal{H}_1$ is true, the channel covariance matrix is

$$\begin{aligned} \mathbf{R} = \mathbb{E}\{\mathbf{h}\mathbf{h}^H\} &= \beta_1 |\mathbf{h}_s^T \mathbf{p}|^2 \mathbf{h}_s \mathbf{h}_s^H + \beta_2 |\mathbf{h}_c^T \mathbf{p}|^2 \mathbf{h}_c \mathbf{h}_c^H \\ &+ \beta_3 ((\mathbf{h}_c^T \mathbf{p}) \mathbf{h}_s + (\mathbf{h}_s^T \mathbf{p}) \mathbf{h}_c) ((\mathbf{h}_c^T \mathbf{p}) \mathbf{h}_s + (\mathbf{h}_s^T \mathbf{p}) \mathbf{h}_c)^H. \end{aligned} \quad (9.105)$$

This matrix consists of three terms, where the first term only utilizes the LOS path while the remaining two terms are created thanks to the reconfigurable surface. Each of the terms has rank one because they are outer products of vectors, but all terms are spanned by $\mathbf{h}_s$ and $\mathbf{h}_c$ so $\mathbf{R}$ has rank two (if $K \geq 2$).

The precoding and surface configuration can be selected to optimize this covariance matrix. The reflection matrix $\mathbf{D}_\psi$ only affects the norm of the cascaded channel vector $\mathbf{h}_c$ in (9.101) since $\mathbf{b}_t^T \mathbf{D}_\psi \mathbf{h}_r$ is a scalar. We showed in Section 9.2 that the magnitude of this term is maximized by (9.27), where the phase-shifts ensure that we sum up N phase-aligned terms. Since $\mathbf{h}_r$ and $\mathbf{b}_t$ are array response vectors where each entry has unit magnitude, it follows that $\mathbf{b}_t^T \mathbf{D}_\psi \mathbf{h}_r = N$ when using the optimal configuration.

The precoding vector should be a linear combination of $\mathbf{h}_s$ and $\mathbf{h}_c = N\mathbf{a}_t$ since these are the two transmission directions that lead to the target. To

maximize the average SNR, we can select the precoding vector that maximizes

$$\begin{aligned}
 \mathbb{E} \{ \|\mathbf{h}\|^2 \} &= \text{tr}(\mathbf{R}) \\
 &= \beta_1 |\mathbf{h}_s^T \mathbf{p}|^2 \|\mathbf{h}_s\|^2 + \beta_2 |\mathbf{h}_c^T \mathbf{p}|^2 \|\mathbf{h}_c\|^2 + \beta_3 \|(\mathbf{h}_c^T \mathbf{p}) \mathbf{h}_s + (\mathbf{h}_s^T \mathbf{p}) \mathbf{h}_c\|^2 \\
 &= \mathbf{p}^H \left((\beta_1 \|\mathbf{h}_s\|^2 + \beta_3 \|\mathbf{h}_c\|^2) \mathbf{h}_s^* \mathbf{h}_s^T + (\beta_2 \|\mathbf{h}_c\|^2 + \beta_3 \|\mathbf{h}_s\|^2) \mathbf{h}_c^* \mathbf{h}_c^T \right. \\
 &\quad \left. + \beta_3 (\mathbf{h}_s^H \mathbf{h}_c) \mathbf{h}_c^* \mathbf{h}_s^T + \beta_3 (\mathbf{h}_c^H \mathbf{h}_s) \mathbf{h}_s^* \mathbf{h}_c^T \right) \mathbf{p}.
 \end{aligned} \tag{9.106}$$

This is a quadratic form with respect to the precoding vector and with a Hermitian matrix in the middle. Hence, it is maximized when $\mathbf{p}$ is selected as the unit-length eigenvector associated with the largest eigenvalue.

With the optimized precoding vector and surface configuration described above, the covariance matrix $\mathbf{R}$ will take a particular value that we will denote as $\bar{\mathbf{R}}$. Based on this matrix, we can derive the Neyman-Pearson detector that gives a desired false alarm probability $P_{\text{FA}} = \alpha$ following the approach from Section 8.3.2. In particular, $\mathbf{y} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2 \mathbf{I}_K)$ under the hypothesis $\mathcal{H}_0$ and $\mathbf{y} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, P\bar{\mathbf{R}} + \sigma^2 \mathbf{I}_K)$ under the hypothesis $\mathcal{H}_1$. Lemma 2.14 says that we should decide on the hypothesis $\mathcal{H}_1$ if

$$\gamma \leq \frac{f_{\mathbf{y}|\mathcal{H}_1}(\mathbf{y}|\mathcal{H}_1)}{f_{\mathbf{y}|\mathcal{H}_0}(\mathbf{y}|\mathcal{H}_0)} = \frac{\frac{1}{\pi^K \det(P\bar{\mathbf{R}} + \sigma^2 \mathbf{I}_K)} e^{-\mathbf{y}^H (P\bar{\mathbf{R}} + \sigma^2 \mathbf{I}_K)^{-1} \mathbf{y}}}{\frac{1}{\pi^K \det(\sigma^2 \mathbf{I}_K)} e^{-\mathbf{y}^H (\sigma^2 \mathbf{I}_K)^{-1} \mathbf{y}}}. \tag{9.107}$$

We can rewrite this condition by using the fact that $\ln(\gamma)$ is a monotonically increasing function for $\gamma \geq 0$:

$$\ln(\gamma) - \ln(b) \leq \sigma^{-2} \mathbf{y}^H \mathbf{y} - \mathbf{y}^H (P\bar{\mathbf{R}} + \sigma^2 \mathbf{I}_K)^{-1} \mathbf{y}, \tag{9.108}$$

where the constant $b = \det(\sigma^2 \mathbf{I}_K) / \det(P\bar{\mathbf{R}} + \sigma^2 \mathbf{I}_K)$ is independent of the received signal $\mathbf{y}$. Hence, the Neyman-Pearson detector decides on $\mathcal{H}_1$ if

$$\|\mathbf{y}\|^2 - \mathbf{y}^H \left(\frac{P}{\sigma^2} \bar{\mathbf{R}} + \mathbf{I}_K \right)^{-1} \mathbf{y} \geq \underbrace{\sigma^2 (\ln(\gamma) - \ln(b))}_{=\gamma'}, \tag{9.109}$$

where γ' is the revised threshold variable that must be selected so that

$$P_{\text{FA}} = \alpha = \int_{\|\mathbf{y}\|^2 - \mathbf{y}^H \left(\frac{P}{\sigma^2} \bar{\mathbf{R}} + \mathbf{I}_K \right)^{-1} \mathbf{y} \geq \gamma'} f_{\mathbf{y}|\mathcal{H}_0}(\mathbf{y}|\mathcal{H}_0) \partial \mathbf{y}. \tag{9.110}$$

The sufficient statistics for target detection is $\|\mathbf{y}\|^2 - \mathbf{y}^H \left(\frac{P}{\sigma^2} \bar{\mathbf{R}} + \mathbf{I}_K \right)^{-1} \mathbf{y}$ and is affected by the precoding and surface configuration through $\bar{\mathbf{R}}$.

Figure 9.24 shows the detection probability, P_D , versus the reference SNR obtained if the radar transceiver has a single antenna and there is no reconfigurable surface. The Neyman-Pearson detector is used with the false

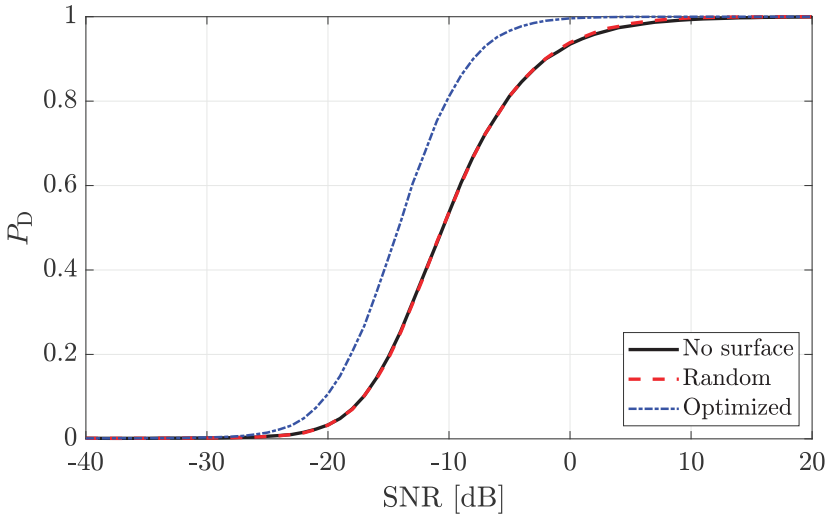


Figure 9.24: The detection probability with respect to the reference SNR in a setup with or without a reconfigurable surface. The surface either has a random phase-shift configuration or is optimized to maximize the received power.

alarm probability $P_{\text{FA}} = \alpha = 10^{-3}$. The radar transceiver is equipped with a half-wavelength-spaced ULA with $K = 10$ transmit and receive antennas. The target location is in the direction $\varphi = 0$ seen from the transceiver, while a reconfigurable surface with $N = 100$ elements is seen in the direction $\varphi = \pi/6$. We let $\beta_1 = \beta_2 N^4 = \beta_3 N^2$ so all the propagation paths are equally strong when the surface is optimally configured. The solid black curve shows the detection performance without the surface, in which case MRT is the optimal precoding. The dashed red curve is obtained when the reconfigurable surface is added to the setup, but it has a random configuration and the precoding still points the signal directly toward the target. The detection probability is improved, but the effect is negligible since the extra paths are weak. It is when the precoding and surface configuration are jointly optimized that we can observe large improvements. The dash-dotted blue curve represents this case and is shifted by roughly 4 dB to the left, compared to the original black curve. This is explained by the fact that total received power $P_{\text{tr}}(\mathbf{R})$ is increased by 3.9 dB. The blue curve is steeper thanks to the spatial diversity gain obtained by having three random RCS coefficients instead of one.

Beyond this basic example, there are many other MIMO radar system configurations and more complex propagation channels where a reconfigurable surface can enhance detection performance. The expected gains are created by ensuring that a larger fraction of the transmitted power reaches the target location and is then reflected toward the receivers, as well as by creating extra spatially distinguishable paths that provide diversity against randomness and improved spatial resolution. We refer to [175], [176] for further details.

9.5 Exercises

Exercise 9.1. Consider the setup from Example 9.4 with equal propagation losses to all elements, such that the end-to-end channel gain is $(\sqrt{\beta_s} + N\sqrt{\beta_r\beta_t})^2$. How many metaatoms N are required for the reconfigurable surface to double the received power, compared to the case of $N = 0$?

- Answer the question when $\beta_s = \beta_t = \beta_r = 10^{-8}$.
- Answer the question when $\beta_s = 10^{-10}$, $\beta_t = 10^{-8}$, and $\beta_r = 10^{-6}$.

Exercise 9.2. The end-to-end channel in (9.28) becomes $h = \sum_{n=1}^N h_{r,n} e^{j\psi_n} h_{t,n}$ if the static channel is totally blocked. Suppose the channels are equally strong to/from all metaatoms: $h_{r,n} = \sqrt{\beta_r}$ and $h_{t,n} = \sqrt{\beta_t}$, $n = 1, \dots, N$.

- What is the average channel gain $\mathbb{E}\{|h|^2\}$ if the phase-shifts ψ_n are selected as independent random variables that are uniformly distributed between 0 and 2π ?
- What will $|h|^2$ become if the phase-shifts are selected to maximize it?
- Compare the results in (a) and (b). What kind of gain is missing in (a)?

Exercise 9.3. The end-to-end channel gain in (9.25) becomes $|\mathbf{h}_r^T \mathbf{D}_\psi \mathbf{h}_t|^2$ if the static channel is totally blocked by some objects (e.g., at high frequencies). Suppose $\mathbf{h}_t \sim \mathcal{N}_C(\mathbf{0}, \beta_t \mathbf{I}_N)$ and $\mathbf{h}_r \sim \mathcal{N}_C(\mathbf{0}, \beta_r \mathbf{I}_N)$ and they are independent.

- What is the average channel gain with a static surface with $\mathbf{D}_\psi = \mathbf{I}_N$?
- What is the average channel gain with a reconfigurable surface that is configured to maximize the channel gain? Hint: $\mathbb{E}\{|h_{t,n}|\} = \sqrt{\beta_t} \sqrt{\pi/4}$.

Exercise 9.4. The LOS end-to-end channel gain is stated in (9.35) when the static channel is negligible. Suppose the transmitter and receiver are equipped with isotropic antennas, while each metaatom has the effective area $A_m = (\lambda/4)^2$.

- Determine an expression of the end-to-end channel gain when the distance between the transmitter and the surface is d_t and the distance between the surface and the receiver is d_r .
- How many metaatoms are needed to achieve an end-to-end channel gain of 10^{-9} if the wavelength is $\lambda = 0.1$ m (i.e., 3 GHz), $d_t = 50$ m, and $d_r = 2$ m? How large is the total area NA_m of the surface?
- How many metaatoms are needed to achieve the same channel gain as in (b) when $\lambda = 0.01$ m (i.e., 30 GHz). How large is the total area NA_m of the surface?

Exercise 9.5. Suppose the reconfigurable surface can turn off specific metaatoms so they absorb all incident signal energy instead of reflecting anything.

- Use this feature to estimate each of the cascaded channels $h_{r,n}h_{t,n}$ sequentially while the remaining $N - 1$ metaatoms are turned off. Follow the ML estimation framework in Section 9.2.2 and assume that $h_s = 0$.
- Suppose all metaatoms are turned on during the ML estimation and simplify the ML estimator in (9.42) for the case when $h_s = 0$.
- Show that the ML estimate can be expressed as $\hat{\mathbf{h}} = \check{\mathbf{h}} + \text{effective noise}$ in both (a) and (b). Compare the variances of the noise terms. Is it preferable to turn metaatoms on/off during the channel estimation? Explain the result.

Exercise 9.6. Consider a reconfigurable surface designed as a uniform planar array with N_H columns and N_V metaatoms per column. Suppose a plane wave impinges from the direction φ_i in the azimuth plane and should be reflected towards a user in the direction φ_o in the azimuth plane (i.e., $\theta_i = \theta_o = 0$).

- Prove that the cascaded channel coefficient $h_{r,n}h_{t,n}$ is equal for all the N_V metaatoms located in the same column.
- Based on the property proved in (a), if we deploy the reconfigurable surface in an environment where signals only propagate in the azimuth plane, we can reduce the number of phase-shift variables from $N_H N_V$ to N_H . This is achieved by assigning the same phase-shift to metaatoms in the same column. Write up the corresponding end-to-end channel and factorize it similarly to (9.38).
- Write up a new ML estimator that utilizes the new factorization from (b). Show that the minimum pilot length is now $L_p = N_H + 1$.
- Suppose the same total energy $(N + 1)q$ is utilized for pilot transmission with the new ML estimator as with one considered in (9.44). How much smaller total variance will the scaled noise term have with the new estimator?

Exercise 9.7. A classic way of extending wireless coverage (e.g., into tunnels) is to use a repeater that picks up the signal using one antenna and immediately retransmits an amplified version using another antenna. In this exercise, it will be compared with a reconfigurable surface in the same deployment scenario.

- The received signal at the repeater is $y_1 = \sqrt{\beta_r}x_1 + n_1$ and the received signal at the receiver is $y_2 = \sqrt{\beta_r}x_2 + n_2$, where $n_1, n_2 \sim \mathcal{N}_C(0, N_0)$. Suppose the data signal $x_1 \sim \mathcal{N}_C(0, q_1)$ is transmitted and that the repeater sends $x_2 = \sqrt{a}y_1$, where a is the amplification gain. How should a be selected to ensure that $\mathbb{E}\{|x_2|^2\} = q_2$?
- What is the SNR at the receiver when using the repeater with the amplification gain obtained in (a)?
- If a reconfigurable surface is used in the same scenario, the SNR would be $qN^2\beta_t\beta_r/N_0$. Derive an expression for how many metaatoms N are required to achieve a larger SNR with the surface than with the repeater.
- Compute the number of metaatoms in (c) if $\beta_t = \beta_r = 10^{-8}$ and $q/N_0 = 10^8$. To make the total transmit power the same in both setups, we let $q_1 = q_2 = q/2$.

Exercise 9.8. The end-to-end channel gain in (9.31) with an optimal phase-shift configuration is $\|\check{\mathbf{h}}\|_1^2$, where the 1-norm is used.

- A MISO channel with the same channel vector achieves the channel gain $\|\check{\mathbf{h}}\|^2$, where the Euclidean norm (2-norm) is used. Which of the two squared norms is the largest? Under which conditions are they equal?
- The phase-shift vector ψ acts as a beamforming vector with $\|\psi\|^2 = N + 1$. If MRT is used with a precoding vector that has the same squared norm, what will be the resulting channel gain? Is it larger than $\|\check{\mathbf{h}}\|_1^2$?

Exercise 9.9. Derive (9.61) from (9.55) step-by-step by utilizing the two properties stated after the equation. Hint: Use the commutative and associative properties of the convolution. The commutative property states that $(f * g)(t) = (g * f)(t)$. The associative property of the convolution is $(f * g * h)(t) = (f * g) * (h)(t) = (f) * (g * h)(t)$.

Exercise 9.10. Consider the reflection coefficient Γ_{0n} when the metaatom's impedance Z_n is given by (9.56).

- (a) What does Γ_{0n} converge to as $f \rightarrow 0$? What is its amplitude and phase?
- (b) What does Γ_{0n} converge to as $f \rightarrow \infty$? What is its amplitude and phase?
- (c) Show that $|\Gamma_{0n}| = 1$ for all f if $R = 0$.

Exercise 9.11. If a known pilot signal is transmitted on all subcarriers, the ML estimator described in Section 9.2.2 can be applied to separately estimate each of the channel vectors $\check{\mathbf{h}}[0], \dots, \check{\mathbf{h}}[S-1]$. However, this is unnecessary because adjacent subcarriers have similar channel vectors.

- (a) Show that $h_\psi[\ell]$ in (9.64) can be expressed as $\boldsymbol{\psi}^T \mathbf{h}[\ell]$, where $\boldsymbol{\psi} = [1, e^{j\psi_1}, \dots, e^{j\psi_N}]^T$.
- (b) Relate $h_\psi[0], \dots, h_\psi[T]$ to $\check{\mathbf{h}}[0], \dots, \check{\mathbf{h}}[S-1]$ using a DFT matrix.
- (c) Use the property in (b) to determine an ML estimator of $\mathbf{h}[0], \dots, \mathbf{h}[T]$, based on the received signals from pilot transmission on $T+1$ subcarriers.

Exercise 9.12. Consider a SIMO channel aided by a reconfigurable surface where the channel vector is

$$\mathbf{h} = \mathbf{h}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{h}_t. \quad (9.111)$$

Suppose there is a free-space LOS channel $\mathbf{H}_r = \mathbf{a}_r \mathbf{b}_r^T$ between the surface and receiver, where $\mathbf{a}_r, \mathbf{b}_r$ are vectors. Determine the surface configuration that maximizes the capacity.

Exercise 9.13. Consider a point-to-point MIMO system with $M = K$ where the channel matrix $\mathbf{H} \in \mathbb{C}^{M \times M}$ has full rank. The system operates at high SNR so that equal power allocation is optimal.

- (a) Prove that the high-SNR capacity is upper bounded by $M \log_2(1 + \frac{q \|\mathbf{H}\|_F^2}{M^2 N_0})$. Under what conditions on $\mathbf{H}$ is the upper bound achieved? Hint: Use the inequality of arithmetic and geometric means from Lemma 3.2.
- (b) The channel contains a reconfigurable surface, so the channel matrix is modeled according to (9.81) as $\mathbf{H} = \mathbf{H}_s + \mathbf{H}_r \mathbf{D}_\psi \mathbf{H}_t$. How should the surface be configured to maximize the high-SNR capacity if $\mathbf{H}_s, \mathbf{H}_r$, and $\mathbf{H}_t$ are rank-one matrices?

Exercise 9.14. Propose an algorithm for downlink sum capacity maximization that switches between solving (6.122) for given user channels and updating the phase-shifts similarly to Algorithm 9.3.